LAND BRANDENBURG

Ministerium des Innern
und für Kommunales

# Large forest fires in the state of Brandenburg

Report on the 2022 wildfire season

State School and Technical Facility for Fire
and Disaster Protection

**Notice**

For the sake of readability, we sometimes refer to individuals or groups of individuals in this report in one form or another, whereby this always refers to both female, diverse and male persons.

Machine Translated by Google

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# List of abbreviations

| | |
|---|---|
| AGBF | Working group of the heads of the professional fire brigades in Brandenburg |
| AFKzV | Committee for Fire Services, Civil Protection and Civil Defense of AK V |
| AK V | Working Group V of the IMK |
| ASBB | Authorized digital radio authority of the state of Brandenburg |
| CHAPTER | Federal motorway |
| BAR | Barnim district |
| BbgBKG | Law on Fire Protection, Assistance, and Disaster Control of the State of Brandenburg (Brandenburg Fire and Disaster Control Act - BbgBKG) |
| BF | professional fire brigade |
| BMI | Federal Ministry of the Interior, Building and Home Affairs |
| BSE | Fire protection unit |
| CBRN | Chemical, biological, radioactive, nuclear |
| DGUV | German statutory accident insurance |
| FROM | German Institute for Standardization |
| DMO | Direct Mode Operation |
| DLRG eV | German Lifesaving Society |
| EE | Elbe-Elster district |
| EU | European Union |
| PROFIT | Command vehicle |
| IN | Euronorm |
| GTLF | Large tank fire engine |
| HuPF | Manufacturing and testing description for universal firefighter protective clothing |
| HVL | Havelland district |
| FF/O | Frankfurt (Oder) |
| IuK | Information and communication |
| IMK | Standing Conference of Interior Ministers and Senators of the States |
| IRLS | Integrated regional control center |
| KampfmV | Regulatory Ordinance for the Prevention of Dangers Caused by Explosive Ordnance (Explosive Ordnance Ordinance for the State of Brandenburg) |
| KatSV | Ordinance on the Units and Facilities of Disaster Protection (Disaster Protection Ordinance - KatSV) |
| KGS | Coordination group of the state government's disaster control staff |
| MOH | Crisis Management Coordination Center of the State Government |
| KMBD | Explosive Ordnance Disposal Service of the State of Brandenburg |
| LFB | Brandenburg State Forestry Authority |
| LDS | Dahme-Spreewald district |
| LFV eV | Brandenburg State Fire Brigade Association |
| LMBV | Lusatian and Central German Mining and Management Company |
| THE | Oder-Spree district |
| LSTE | State School and Technical Facility for Fire and Disaster Protection of the State of Brandenburg |

| | |
|---|---|
| LWaldG | Forest Act of the State of Brandenburg |
| MdF | Ministry of Finance of the State of Brandenburg |
| WITH | Modular deployment unit |
| WHAT | Ministry of the Interior and Local Government of the State of Brandenburg |
| MLUL | Ministry of Rural Development, Environment and Agriculture of the State of Brandenburg |
| MoFüst | Mobile command support staffs |
| MOL | Märkisch-Oderland district |
| OHV | Oberhavel district |
| PM | Potsdam-Mittelmark district |
| PSA | Personal protective equipment |
| SPN | Spree-Neisse district |
| TEL | Technical Operations Management |
| TF | Teltow-Fläming district |
| THW | Federal Agency for Technical Relief |
| TEL | Tank fire engine |
| WaldBrSchVO MV | Ordinance on the prevention and control of forest fires (Forest Fire Protection Ordinance - WaldBrSchVO) of the state of Mecklenburg-Western Pomerania |
| ZDPol | Central Police Service of the State of Brandenburg |

# glossary

**Windward**

windward side

**Crown closure degree**

Stock density

| Final degree | Characteristic |
|---|---|
| crowded | The crowns interlock deeply. |
| closed loose | The crowns touch the tips of the branches. |
| | The crown spacing is smaller than one crown width. |
| light | The crown spacing corresponds to one crown width. |
| spacious | The crown spacing exceeds one crown width. |

**calamity**

Severe damage caused by pests, hail, storms, etc.

**Calamity logging**

Wood that has been felled and is available for further use as a result of storm damage, drought, pest infestation, etc.

**Stock umbrella**

existing tree population

**Convergence line**

Areas of converging air near the ground. The winds that occur here always flow from the area of higher air pressure to the area of lower pressure.

**pre-clinical emergency medicine**

Emergency medical services, medical assistance for emergency patients outside a hospital

**Curricula**

Curricula

**Fleetmapping**

Technical summary of several voice subscribers or call groups

**Tax cracks**

Loading rack/Loading rucksack

**Anthropologie**

Human Science

# 1. Introduction

Forest and vegetation fires are not unknown to the population, emergency services, and specialists in the state of Brandenburg. With its large forest area, the prevailing subcontinental climate, and the light sandy soils, the state of Brandenburg is one of the most vulnerable regions in Germany.
With low water retention capacity and the dominant tree species pine (Pinus sylvestris), it is one of the areas most prone to forest fires in Germany. An additional special feature of the state of Brandenburg is that of a total of approximately 1 million hectares of forest area, about 292,000 hectares are suspected of containing explosive ordnance. In the event of a fire, this complicates ground-based firefighting in the affected areas and, in individual cases, requires aerial firefighting support from the Federal Police or Bundeswehr. Given climate change and the prevailing conditions in the country that favor forest fires (pine density, site conditions, munitions contamination), the risk of forest fires is increasing and must be reduced. Preventive forest fire protection measures primarily target forest restructuring, limit conditions that favor fire outbreaks, and create improved conditions for successful forest firefighting.

While in the past decade, citizens and emergency services have regularly been aware of smaller forest fires with moderate destructive potential, this perception has changed significantly since 2018. The year 2022 in particular has shown all stakeholders the fire risk that can arise from persistent high pressure, combined with a lack of precipitation, prolonged sunshine intensity and strong

winds, on vegetation areas. This circumstance was further aggravated by insignificant increases in the proportion of calamity wood on the forest areas (winter storms in 2022, drought stress, insect infestation).

In addition to a large number of moderate forest fires that were brought under control within a short period of time by the local fire departments, a few forest fires developed extreme fire behavior due to unfavorable conditions. In some cases, several hundred hectares of forest were destroyed within a few hours due to very high propagation and spreading speeds. During the high,

Phases of fire intensity that are difficult to control also lead to direct (danger of fire spreading) or indirect (due to smoke) threats to residential areas, resulting in evacuation orders for residents. The energy release phenomena of these few, extreme forest fire events indicate a future trend and were previously unknown to the majority of emergency responders in this form. The population living in the vicinity of the forest fires had understandable concerns for their safety and property.

Furthermore, in addition to pure forest fire events, fires[1] on agricultural land, wasteland, and other open land[2] must also be included in the scope of this report. In particular, the existing interconnection between settlements and contiguous vegetation areas[3] can result in a greater risk impact[4] in wildfires. The resulting dangers and damage must be prevented.

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*1Vegetation fires as a generic term for all types of fires on vegetation areas*

*2Open land areas are areas that are not built up or dominated by trees, thus all biotope types that are not Counting the forest*

*3Wildland-Urban-Interface*

*4Risk = probability of occurrence x extent of damage*

or at least to reduce it by using the knowledge gained this season to
Potential for optimization and further development is identified. In addition to sustainable preventive threat mitigation measures, the focus is also on further improving defensive structures. The basis for this report is:

- the information and documents available to Department 34 (MIK) and the LSTE
- the presentations and discussion points of the evaluation event on the forest fire events in 2022 of the LSTE on 27 October 2022 in Lübbenau/Spreewald
- the discussion results of the advanced training seminar for district fire chiefs and managers of the professional fire brigades from 24 to 25 November 2022
- Findings of the interdepartmental forest fire working group in the MIK with the participation of district fire chiefs and interest groups (LFV BB eV, AGBF BB)
- The statistical surveys of the Brandenburg State Forestry Authority

The emergency services performed intensive and excellent work over several weeks/months.

# 2. Special forest fire events in 2022

The 2022 season was with 521 forest fires and a damaged forest area of 1,425.50ha5 an exceptional year. Larger damages were last recorded in 1983 and 2018 The special feature of the past year was the dimension of the Forest fires. Eight fires were larger than 10 hectares, four of which were larger than 100 hectares.

More than half of all forest fire reports were at forest fire danger levels 4 and 5. At the same time, these forest fire reports caused 98% (1,380 ha) of the total seasonal damage area (see Figures 1 and 2).

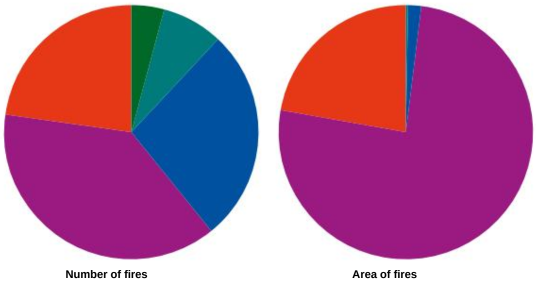


Figure 1: Graphic representation of fire alarms and areas, WGSt dependent (Source: LFB, 2022)

Legend/Value table

2022 Total fires: 507

| | Level | n | n% | Area | In% |
|---|---|---|---|---|---|
| | | 22 | 4,34 | 1,64 | 0,12 |
| | 1 | 39 | 7,69 | 5,98 | 0,42 |
| | 2 | 138 | 27,22 | 23,37 | 1,66 |
| | 3 | 193 | 38,07 | 1.068,97 | 75,76 |
| | 4 5 | 115 | 22,68 | 310,94 | 22,04 |

Figure 2: Numerical and percentage breakdown of forest fire reports/ burned areas, depending on the WGS (Source: LFB, 2022)

The appropriate non-police, special hazard prevention potential was not sufficient in all operational situations exclusively at the level of the responsible authorities for local fire protection and local assistance, so that in some fires the elements of a major damage event according to Section 1 Paragraph 2 No. 1 of the Law on Fire Protection, Assistance and Disaster Control of the State of Brandenburg (BbgBKG) were met.
The responsible authorities for supra-local fire protection and supra-local assistance or the lower civil protection authorities subsequently took over the last season the responsibility for hazard prevention. As Figure 3 clearly shows, the majority of forest fires (30 out of a total of 38) that were classified as at least major damage events were concentrated in the most recent observation period from 2018 to 2022.

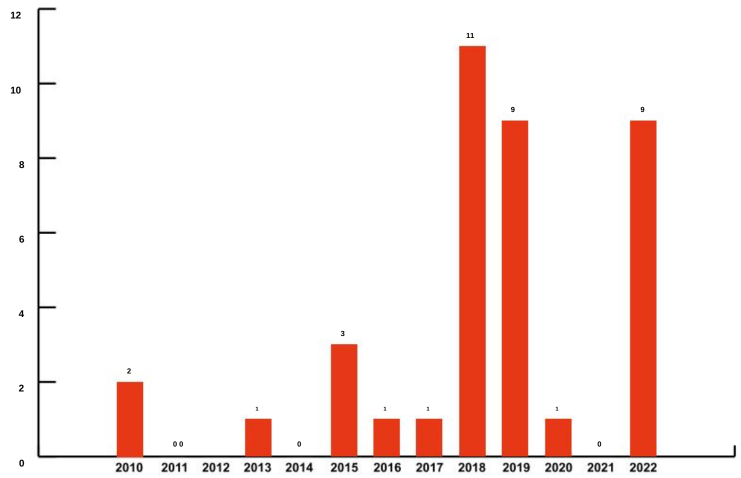


Figure 3: Number of forest fires > 10 ha. Usually, therefore, the status is a major damage event (Source: LFB, 2022)

5Excluding the federal state, there were a total of 507 forest fires on 1410.90 hectares.

The major operations in the 2022 season were operations that mostly involved
lasted for several days, sometimes posed a direct threat to or even affected settlement areas, and required a high level of supra-local and supra-regional forces and resources for ground and/or air-based damage control.

| local authority | district | Beginn | End | destroyed Area (Forest)6 | cross-federal and cross-state | Suspicion of explosive devices | Settlement areas7 | Eviction/ Evacuation-rung |
|---|---|---|---|---|---|---|---|---|
| Treuenbrietzen | PM | 17.06.2022 1:14 p.m. | 23.06.2022 173 | | no | and | threatened | and |
| Beelitz | PM | 19.06.2022 12:15 p.m. | 22.06.2022 233 | | no | and | threatened | and |
| Liebenwerda (Local communities of Bad Liebenwerda & Mühlberg/Elbe) | EE | 23.06.2022 3:47 PM | 29.06.2022 316 | | yes (SN) | and | threatened | and |
| Lieberose/Oberspreewald LDS | | 04.07.2022 9:36 a.m. | 26.07.20228 90 | | no | and | no | no |
| Liebenwerda (local community Falkenberg/ Elster) | EE | 25.07.2022 13:34 | 02.08.2022 422 | | yes (SN) | and | affected | and |

*Table 1: List of major damage events caused by vegetation fires in 2022.*

From the list in Table 1 it is clear that the areas of all listed major damage events were classified as areas suspected of containing explosive ordnance or even individual explosive ordnance
This required compliance with safety distances9 in accordance with
Fire Service Regulation 500 (units in ABC operations) and DGUV Information 201-027 as well as the consultation of the expert advice of the Explosive Ordnance Disposal Service of the Central Service of the Brandenburg State Police (KMBD). Although safety distances according to
Since the fires could be adapted to the respective assessment based on expert advice, direct firefighting was generally not possible. Further significant vegetation damage had to be accepted as part of the balancing of interests. The use of special technology (armored
Clearing equipment of the Bundeswehr, circular sprinklers, etc.) had to be arranged.

Compared to forest fires in recent years, there is an increasing threat to residential areas. As a necessary consequence, clearance and evacuation measures had to be at least prepared or even implemented. Direct firefighting on the burnt areas became less important in light of such dynamic changes in the hazards. Instead, a large portion of the forces and resources on site were diverted to
Protection of residents and settlement areas.

*6The affected agricultural land and other burned areas (e.g. wasteland, set-aside areas) were not recorded statistically. Depending on the event, several tens to hundreds of hectares of these to accept areas.*

*7Settlement areas include areas with a building character (primarily residential areas, industrial and commercial areas, Mixed-use areas and areas with a special functional character) excluding traffic and water areas (cf. https://www.ioer-monitor.de/methodik/glossar/s/siedlungsflaeche/, accessed 07.12.2022).*

*8The major disaster situation ended on 9 July 2022 at 10:00 a.m. Until 27 June 2022, at least 2 Tank fire engines from local authorities in LDS are in operation for 24 hours each, as high Temperatures in the moorland area caused embers to flare up again.*

*9FwDV 500 in the version of January 2012 according to measure group 1: Danger zone: 500 m, cordoned-off area: 1,000 m (currently valid FwDV 500 introduced in BB; amended FwDV 500 in the version of January 2022 with the same distance values).*

Due to the alarm and arrival times, units were only available for further measures with a delay.

It can also be seen that the majority of forest fires were reported between early and late afternoon. Figure 4 illustrates this by considering all 507 forest fire reports throughout the day. Daily maximum temperatures are also regularly reached during this time window. Due to the high position of the sun and
Their intensity – under clear or moderately cloudy skies – heats combustible materials and removes residual moisture. The availability of combustible material for a continuous fire process increases.

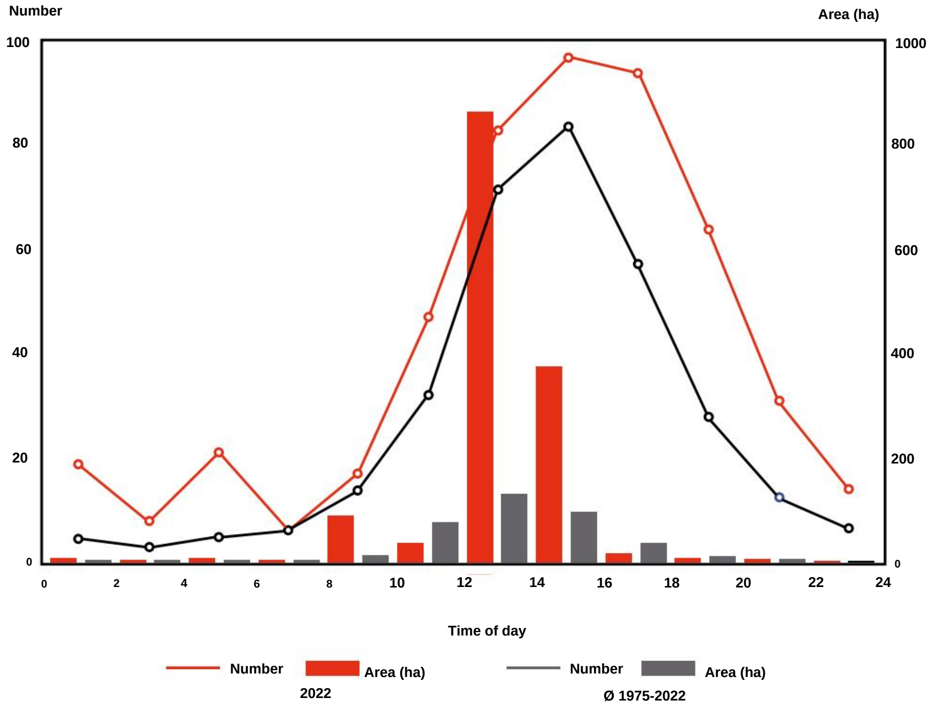


*Figure 4: Daily distribution of forest fires (Source: LFB, 2022)*

### 2.1. Forest fire near Treuenbrietzen OT Frohnsdorf

On June 17, 2022, a vegetation fire broke out in the Treuenbrietzen district near the Frohnsdorf district on a forest area that had already burned in 2018. Part of the former burnt area is the subject of research in the PYROPHOB project. The proportion of deadwood was very high at the time of the fire. The lack of a canopy made the open
Area more susceptible to wind influences.

#### Location description

Due to the known contamination of explosive ordnance, the first-aided forces limited themselves to defensive measures from a safe distance. The necessary resources required supra-local support, so the Potsdam-Mittelmark district took over command of the operation in the Frohnsdorf district that afternoon and treated the situation as a major incident. The fire was contained to the area of the fire of 2018 until the afternoon of June 18, 2022. Along the paths around the initially affected area,

Forestry technology was used to remove standing and lying dead wood from the burned area in order to
To minimize fire bridges to unaffected stocks and to ensure safety at the site

Due to the existing explosive ordnance and the continued high burning rate of the standing deadwood trunks, the emergency services were at high risk, making direct follow-up firefighting operations impossible or only possible to a limited extent.

Due to shifting and increasing winds, the wildfire spread on the afternoon of June 18, 2022 to the adjacent, previously unaffected forest area. Direct firefighting was not possible here either due to the suspicion of explosive ordnance. The fire front moved towards the B 102 federal highway and the Tiefenbrunnen district. Subsequently, blocking positions were set up along the B 102 federal highway and the evacuation of Tiefenbrunnen was prepared. Helicopters supported the containment measures. On the evening of June 18, 2022, the district administrator declared a state of emergency. During the night, the situation calmed slightly. From the following morning until noon,

the wind picked up again and increased significantly, so that further fire spread along the B 102 in the direction of Klausdorf, Frohnsdorf and Tiefenbrunnen had to be determined.
The evacuations of the endangered villages and communities were initiated. All available units from the district were deployed during this phase, maintaining basic security. Not all fire protection units that had already been requested on the morning of June 19, 2022, could be deployed in the Frohnsdorf area, but had to be diverted due to another large forest fire in the Potsdam-Mittelmark district (see forest fire in Beelitz). Later on June 19, 2022, the situation stabilized due to decreasing wind and light precipitation. The burned forest area was cleared by

The Brandenburg State Forestry Authority estimates the area to be 173 hectares.

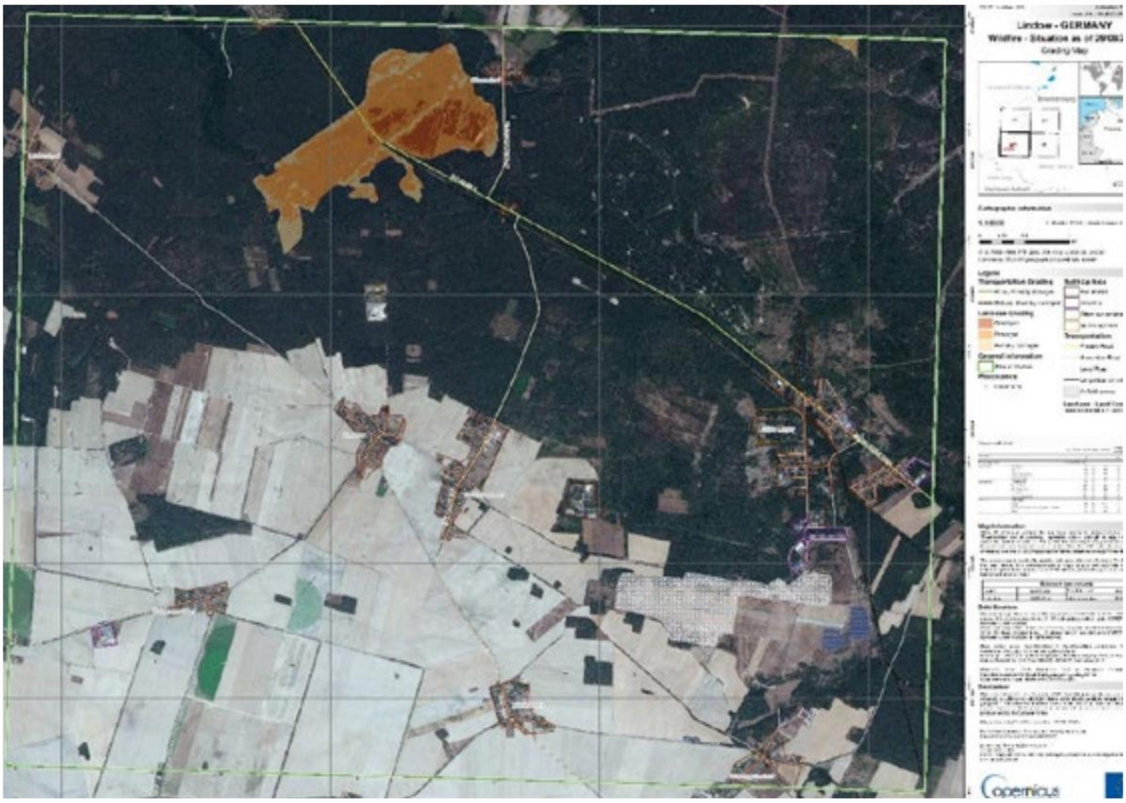


*Fig. 5: Destroyed forest area Frohnsdorf 2018; Source: Copernicus Emergency Management Service (© 2018 European Union), EMSR307*

**Forces and resources deployed**

During the acute phases of the Frohnsdorf forest fire, approximately 440 emergency personnel were deployed simultaneously. Among those deployed were:

- Local fire brigades from the Potsdam-Mittelmark district, partly as pre-structured, supra-local units (TLF platoons) for the defined support of a local authority within the district

- Disaster control units of the Potsdam-Mittelmark district: Fire protection unit (BSE), guided tour with SEG-Fü, SEG catering (SEG-V), SEG care (SEG-Bt), emergency pastoral care

- Disaster relief units (BSE, leadership, SEG-V, SEG-water hazards) from other Brandenburg districts/independent cities:
  - City of Potsdam
  - City of Brandenburg an der Havel
  - Havelland district
  - Elbe-Elster district
  - Teltow-Fläming district
  - Dahme-Spreewald district
  - Ostprignitz-Ruppin district
- State police with:
  - Guard and shift duty
  - Police helicopter squadron for aerial situation assessment
  - Water cannons of the technical operations unit
  - Explosive ordnance disposal service
- LSTE and Brandenburg State Forestry Authority
- Emergency pastoral care
- Federal Police with:
  - Helicopters (including external fire water tanks)
  - Coordination and supply resources for helicopter operations
- Bundeswehr with:
  - District Liaison Command
  - Helicopter (including external fire water tanks)
  - Coordination resources for helicopter deployment
  - armored clearing technology
- THW with:

- FGr Emergency Supply and Emergency Repair

- Command units

• other external service providers and non-profit aid organizations:

- agricultural and forestry contractors

- @fire - International Disaster Relief Germany e. V.

- “Disaster Recovery Management” of Deutsche Telekom AG

• MIK Crisis Management Coordination Center

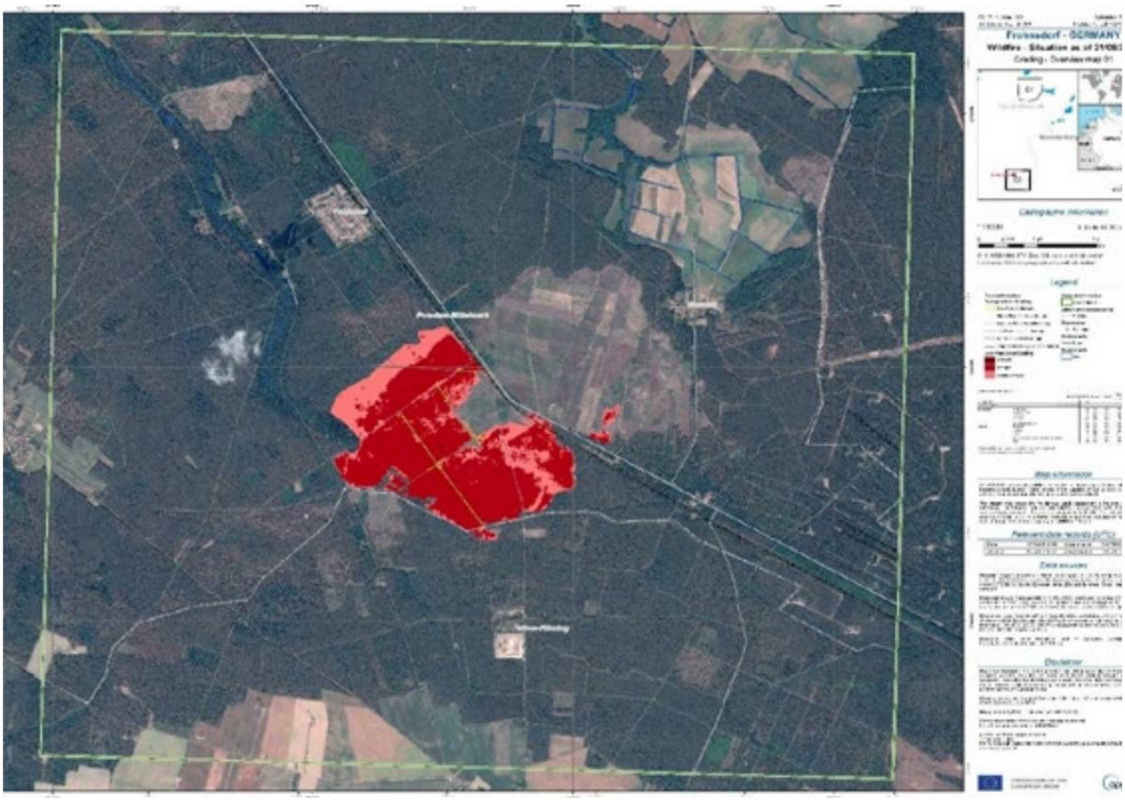

*Fig. 6: Frohnsdorf fire area 2022; Source: Copernicus Emergency Management Service (© 2022 European Union), EMSR582*

**Special features**

During the operation near Frohnsdorf, the following special features and circumstances were observed, which influenced the course of the operation:

• unfavourable weather-related influences with very high outside temperatures (> 30°C), low relative humidity (around 30%) and partly gusty, shifting winds (gusts over 30 km/h)

• Strong spot fire phenomena (flying sparks and flying sparks over the Jüterbog railway line Treuenbrietzen and the B 102)

• Endangerment of settlement areas including evacuation orders for several Local and municipal districts

• Communication disruptions in digital radio BOS (radio cell overload due to avoidable data traffic) and mobile communications (underserved areas)

- phased high speeds of the fire front
- Suspicion of explosive ordnance or contamination (during the operation several ammunition bodies of different calibers perceived on the surface)
- Airborne support of firefighting by means of water extraction from open Waters (here: quarry lake in Treuenbrietzen):
  - Securing the water body with SEG water hazards
  - The process of requesting the Bundeswehr's airborne capability was complicated by delayed media coverage (media report from midday on June 18, 2022: Fire is under control). The Bundeswehr's decision-making body questioned the Bundeswehr's need for subsidiary support, despite the acute change in the situation.
- Parceling of the fire area and endangered forest areas by creating clearings with armored clearing technology of the Bundeswehr
- further forest fires occurring in the vicinity of the large forest fire:
  - 18 June 2022: in the Lüdendorf/Lindow area
  - June 19, 2022: in the Rietz/Nichel and Beelitz area (developed into another large forest fire in the Potsdam-Mittelmark district, see 2.2)
  - Operations had to be coordinated by the Technical Operations Management in Frohnsdorf
  - causal connection with large forest fire (flying sparks, flying fire) can be excluded
- Declared a state of emergency on June 18, 2022, at 6:45 p.m. by the District Administrator (so far the first state of emergency due to a forest fire in the state of Brandenburg since reunification).

**2.2. Forest fire near Neuseddin/Beelitz**

During the response to the Frohnsdorf forest fire, another forest fire broke out in the Potsdam-Mittelmark district west of the Seddin marshalling yard at noon on June 19, 2022, under weather conditions favorable for forest fires. Numerous local units from the district were already involved in the Frohnsdorf operation. Due to the fire's origin northwest of a railway infrastructure crossing, both the precise location and the exploration of suitable access routes were difficult and time-consuming. These conditions favored both the spatial and intensifying spread of the fire.

**Location description**

The first-alarmed emergency services from the municipality of Seddiner See and the city of Beelitz managed
Due to its advanced extent, it was not possible to control the fire and prevent it from spreading in a southerly direction
across the Dessau railway line (east-west route).
prevent. Wind gusts of around 60 km/h led to enormous speeds of the fire front,

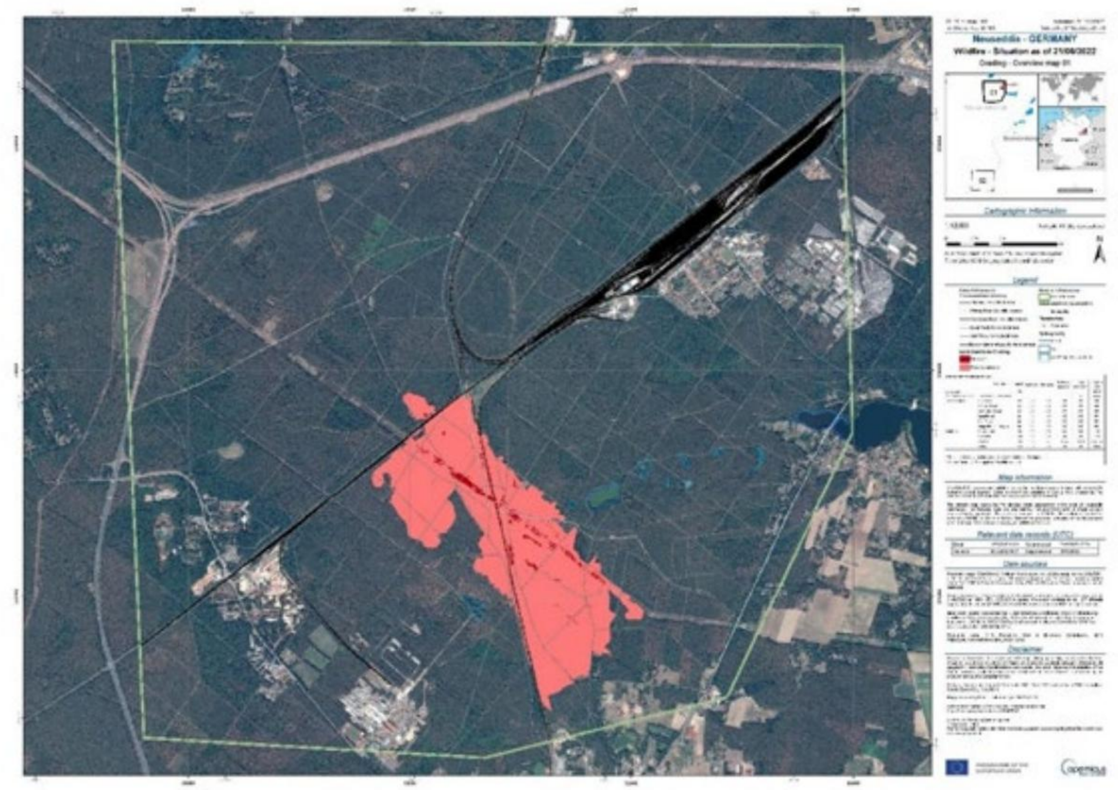

*Fig. 7: Fire area Neuseddin 2022; Source: Copernicus Emergency Management Service (© 2022 European Union), EMSR582*

which at times reached up to several hundred meters per minute. As a result, residential areas of Beelitz-Heilstätten and Beelitz, as well as the Hans-Joachim-von-Zieten barracks of the German Armed Forces, were endangered. Direct firefighting of the fire perimeter was impossible due to long detours and the maintenance of safety distances in some areas.
(Perception of detonation noises) could only be implemented with a delay. The fire ultimately also spread over the north-south railway line Berlin-Jüterbog. Presumably
The main wind direction from the northwest and the
A low-vegetation energy route running from northwest to southeast. Both the corridor
The 380 kV high-voltage overhead line and the Berlin-Jüterbog railway line both facilitated the spread in a southeasterly direction due to the incision in the vegetation profile10 and acted as a flow path due to the parallel wind. A nearby
The substation, which provides energy to the surrounding settlements, was in immediate danger.
The firefighters managed to contain the fire flank directly at the perimeter.
to secure the substation and defend the facility. Additional fire protection units, already requested for the forest fire in Frohnsdorf, were diverted to the Beelitz operational area during the march and supported the local emergency services during the acute afternoon and evening hours. Due to the growing command organization, a technical operations command was set up in the fire station of the Beelitz local fire department and supported by the command unit of the city's professional fire department, which was requested via the KKM.
Potsdam occupied. During this time, the fire front extended over a distance of approximately

*10no existing shielding connection under the overhead line or the railway infrastructure to the adjacent forest in-nenrands*

500 m and now endangered a bungalow settlement, the properties of a chicken egg producer, and several single-family homes. The inhabited properties in the danger zone were evacuated by the Beelitz city administration. In addition, a property defense was initiated by the construction facilities in this area.

Due to the dimensions of the fire area, the time-critical distance reduction between the fire front and the first settlement areas (approx. 100 m) as well as the disproportion of Due to the size of the deployment site, a tactical fire operation was initiated in front of the endangered settlement areas on the night of 19 to 20 June 2022. The tactical objective The reduction of fire intensity in the front area and the timely securing of the The tactical fire operation ignited flammable organic materials in front of the front line in a controlled manner, allowing the advancing fire front to flow into this targeted burnt-out area and contain it. This measure secured the settlement before the onset of rain, which finally brought the affected forest fire area to approximately 233 hectares.

**Forces and resources deployed**

During the rapid expansion phase, up to 660 forces were deployed simultaneously:

- Local fire brigades from the Potsdam-Mittelmark district
- Disaster relief units (BSE, leadership, SEG-V, SEG-water hazards) from other Brandenburg districts/independent cities:
  - City of Potsdam
  - Havelland district
  - Oder-Spree district
  - AGBF command unit
- Disaster relief units from the state of Berlin and the state of Saxony-Anhalt
- State police with:
  - Guard and shift duty
  - Police helicopter squadron for aerial situation assessment
  - Explosive ordnance disposal service
- LSTE and Brandenburg State Forestry Authority
- Emergency pastoral care
- Bundeswehr with:
  - District Liaison Command
  - Helicopter (including external fire water tanks)
  - Coordination resources for helicopter deployment
  - Relocation of a firefighting contingent of the German Federal Armed Forces (on the way to ILA coverage)
  - armored clearing technology

- THW with:
  - FGr Emergency Supply and Emergency Repair
  - Command units
- other external service providers and non-profit aid organizations:
  - agricultural and forestry contractors
  - @fire - International Disaster Relief Germany e. V.
  - Flughafengesellschaft Berlin-Brandenburg GmbH with 2 airfield fire engines
- MIK Crisis Management Coordination Center

**Special features**

- In connection with the operation at Neuseddin/Beelitz, the following knowledge was gathered and special measures were implemented:
- unfavorable weather-related influences with very high outside temperatures (approx. 35°C), low relative humidity (approx. 25%) and sometimes gusty, shifting winds (gusts over 50 km/h)
- Running speeds of 300 m/min were observed in the fire front at certain times (Note: this value corresponds to the scientifically proven rule of thumb[11]

  **in Lauf=0.1 xv $_{\text{Wind}}$ [m/s])**
- Airborne support of firefighting by means of water extraction from open bodies of water
- Suspicion of explosive devices
- Parceling of the fire area and endangered forest areas by creating clearings with armored clearing technology of the Bundeswehr
- The District Administrator had already declared a disaster the day before due to the incident at Frohnsdorf proclaimed:
  - disaster control units requested for Frohnsdorf were partly diverted on the way to Beelitz
  - Helicopters deployed in Frohnsdorf were withdrawn to Beelitz
- Tactical fire deployment to quickly and effectively reduce fire intensity and -spread
- Establishment of a staging area in the Hans-Joachim-von-Zieten barracks (one-finally, provision of food through the military kitchen)
- Adaptation of the management organization: Formation of a disaster management team by Potsdam-Mittelmark District Administration as the new final authority due to the two installed technical operations management teams:

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*11 CRUZ, M.G., ALEXANDER, M.E. The 10% wind speed rule of thumb for estimating a wildfire's forward rate of spread in forests and shrublands. Annals of Forest Science 76, 44 (2019), page 2. https://annforsci.biomedcentral.com/articles/10.1007/s13595-019-0829-8*

- The performance potential of local fire protection and supra-local disaster prevention was significantly overstretched by the two ad-hoc situations despite extensive human and material resources (rest periods for deployed forces, commitment of own management personnel in several command posts)
- The district considered transferring the operational command to the state. moved

**March 2nd forest fire near Bad Liebenwerda and Mühlberg/Elbe**

On the afternoon of June 23, 2022, the fire department of the Liebenwerda municipality, located on the Brandenburg-Saxony border, received a report of a forest fire in the Kröbeln district (Bad Liebenwerda municipality). After extensive investigation, the fire was localized to the site of the former Zeithain military training area on the Saxon side. A significant, verifiable contamination of explosive ordnance was detected.
made targeted, early intervention difficult.

**Location description**

The spread of the fire on this day was initially limited to the Saxon side
on 32 hectares. In coordination with the Saxon operational command, the local forces of the association of municipalities secured the Altenau district along the north-south Falkenberg/Elster - Riesa railway line with a barrier during the evening and night hours. During the late morning of June 24, 2022, a convergence line with local severe weather potential crossed the operational area. With an outside temperature of around 30°C,
very dry air masses near the ground and a freshening, gusty wind, this
Weather phenomenon for the so far moderately spreading forest fire, a so-called ground fire, sufficient potential for revived, extreme fire behavior.
Due to the lack of precipitation, the fire spread from a ground blaze to a partial full-blown fire, primarily spreading north. Due to the extreme fire dynamics, the firefighters, who were reinforced over time, withdrew from the forest area and established a defensive line along the northern border of the Gohrischheide nature reserve.
Neighbouring grain fields were harvested or slashed to prevent the sparks from spreading further. The fire spread across the
A defense line up to 80 meters deep was destroyed by the strong gusts of wind. Further forest and agricultural areas were set ablaze. Emergency personnel withdrew once again and began securing the structures of the now threatened district of Kosilenzien (in the municipality of Bad Liebenwerda), northeast of the fire, and initiating the evacuation of the buildings.
The Elbe-Elster district administration was informed of the danger and damage situation and classified the operation as a major incident as of 2:30 p.m. The Neuburxdorf fire station became the headquarters of the stationary technical operations command. The administrative staff was
The evacuation of the Kröbeln district was prepared, but not carried out. Further significant spread to forest and
Grain fields west of the railway line had to be restricted by supra-local forces in order to reduce the dangers for the villages of Neuburxdorf-Siedlung, Wendisch-Borschütz and
Altenau. Only during the night of June 24, 2022, was the situation brought under control and the forest fire area limited to approximately 316 hectares. A further approximately 80 hectares of open space (grain, fallow land) fell victim to the flames. Ultimately, a fire to the south

The fire spread to buildings in a cornfield adjacent to the village of Kosilenzien.

By the evening of 27 June 2022, the fire on the Saxon side had still not been extinguished.
be brought under control. Forces and resources from the Elbe-Elster district, which were involved in the firefighting operations on Brandenburg territory, were sent to the community of Zeithain to provide support. During this intensive resurgence of the

The fire once again poses a threat to the Brandenburg region. Additional forces from the Elbe-Elster district were also alerted to this incident. A grain field fire that was developing at the same time near the Jacobsthal district (Zeithain municipality) was dispatched with additional units and reserve firefighters from the Liebenwerda municipality, as almost all local fire departments in the Zeithain municipality were tied up by the large forest fire. The rapid intervention eliminated the acute threat to this district.

**Forces and resources deployed**

During the intensive implementation of measures, up to 150 emergency personnel were deployed simultaneously. The following forces and resources were deployed on the Brandenburg side:

- Local fire brigades from the Elbe-Elster district
- Disaster relief units (BSE, leadership, SEG-V, SEE-San) as well as situationally assembled platoons from the Elbe-Elster district and other Brandenburg districts/independent cities:
  - City of Potsdam
  - Dahme-Spreewald district
  - Oberspreewald-Lausitz district
  - Spree-Neisse district
  - Oder-Spree district
  - Barnim district
  - City of Cottbus
- State police with:
  - Guard and shift duty
  - Police helicopter squadron for aerial situation assessment
  - Explosive ordnance disposal service
- Emergency pastoral care
- Bundeswehr with:
  - District Liaison Command
  - Bundeswehr Fire Brigade
- LSTE and Brandenburg State Forestry Authority

• THW with:

- FGr Emergency Supply and Emergency Repair
- Command units

• other external service providers, non-profit aid organizations and special units:

- agricultural and forestry contractors
- @fire - International Disaster Relief Germany e. V.
- Waldbrandteam e.V.
- SKV - Specialized vegetation fire forces (intercommunal cooperation and Specialization of the towns of Zehdenick [OHV], Liebenwalde [OHV] and Templin [UM])

• MIK Crisis Management Coordination Center

**Special features**

In connection with the operation at Bad Liebenwerda and Mühlberg/Elbe, the following knowledge was gathered and special measures were implemented:

• unfavourable weather-related influences with high outside temperatures (approx. 30°C), low relative humidity (approx. 22%) and partly gusty, shifting winds (gusts over 60 km/h)

• In phases, running speeds of 300 m/min were observed in the fire front (Note: this value corresponds to the scientifically proven rule of thumb11 $v_{run} = 0.1 \times v_{Wind}$ [m/s])

• Documentation of local tornadoes/small tornadoes including burning Components (spot fire danger beyond defined defense lines)

• Ad hoc assembly of inter-municipal trains with tank fire engines (from several southern Brandenburg districts) by the Lausitz Regional Control Center to bridge the arrival times of the BSE (afternoon/evening of June 24, 2022)

• Suspicion/confirmation of explosive ordnance

• Use of circular sprinklers to create an unmanned locking position

• Evacuation order in the OT Kosilenzien, evacuation preparation in the OF Kröbeln

• Relocation of a Saxon command vehicle 1 (ELW 1) to the technical operations management in Neuburxdorf with the aim of direct exchange of information on situations and measures between the operations management in Brandenburg and Saxony

• Activation of the administrative staff of the Elbe-Elster district administration

• Carrying out intensive, manual extinguishing work on a fire area (approx. 22 ha) near the Kosilenzien subdivision due to the thickness of the burnt-through raw humus layer and the potential risk of re-ignition (including endangerment of the subdivision) ÿ first joint operation by @fire, Waldbrandteam eV and SKV as well as practical knowledge transfer with volunteer emergency services

## 2.4. Forest fire near Lieberose OT Butzen

Within the Schwielochsee district of the Lieberose/Oberspreewald district,
In the morning hours of July 4, 2022, a forest fire broke out on the area of the Natural Landscape Foundation Brandenburg. Due to the suspicion of explosive ordnance (part of the former military training area Lieberoser Heide) and the involved moorland area "Großes Zehme",
It is not possible to approach the edge of the fire directly and thus fight the fire directly.

### Location description

The local fire department units of the district were initially alerted to the scene of the incident in accordance with the alarm and dispatch regulations, assembled at a designated staging area, and conducted a comprehensive reconnaissance of the area. The fire area with slow to moderate spread was located within an area classified as suspected of containing explosive ordnance. The fire department leadership and the district fire chief of the Dahme-Spreewald district decided to prioritize defense measures for unaffected wilderness areas over the intact
There was a direct threat to residential areas
At no time during the operation was this known. Given the limited technical and tactical capabilities of the district's volunteer fire department, supra-local command and firefighting support was called in, and the operation was classified as a major disaster. Subsequently, the fire spread to other moorland areas and adjacent forest areas.
In inaccessible areas (no developed forest path system), the firefighters slowly moved northward against the wind. On July 5, 2022, the command center decided to use reconnaissance paths that were sufficiently far from the moving fire edge for a blocking position with circular sprinklers. Open standing waters in the indirect
Stable water intake points were provided nearby. Three centrally procured high-performance delivery systems (HFS) secured the water requirements for the circular sprinklers over a delivery distance of approximately 6 km, which were positioned around several circumferential sides over a total distance of more than 4 km.
The advantage of the circular sprinkler barrier was particularly appreciated in its unmanned and continuous water release. Additional sections were regularly patrolled with water-carrying vehicles.
In addition, airborne firefighting measures were initiated for several days to support the fire. In inaccessible areas, the fire edge thus ran towards the wet area. The measures led to a safe stagnation of the fire spread and limited the damage event to a

An area of 95 hectares was affected by light rainfall. Despite the handling of the major disaster on July 10, 2022, and the handover of the site to the local authority, additional local authorities from the entire Dahme-Spreewald district were permanently deployed in shifts to support the office for weeks.
The fire was handed over to the landowner on July 26, 2022. Until August 16, 2022, further alarms were issued irregularly within the office due to ember pockets that continually reignited.

### Forces and resources deployed

Up to 400 emergency personnel were deployed simultaneously in the area. The following forces and resources were deployed, among others:

- Local fire brigades from the Dahme-Spreewald district

• Disaster relief units (BSE, leadership, SEG-V) as well as situationally assembled
Trains from the Dahme-Spreewald district and other Brandenburg districts/independent cities:

- City of Cottbus & City of Potsdam
- Märkisch-Oderland district
- Barnim district
- Potsdam-Mittelmark district
- Teltow-Fläming district
- Oberspreewald-Lausitz district
- Spree-Neisse district
- Oder-Spree district
- Barnim district

• State police with:
- Guard and shift duty
- Police helicopter squadron for aerial situation assessment
- Explosive ordnance disposal service

• Federal Police with:
- Helicopters (including external fire water tanks)
- Coordination and supply resources for helicopter operations

• Bundeswehr with:
- District Liaison Command

• LSTE and Brandenburg State Forestry Authority

• THW with:
- FGr Emergency Supply and Emergency Repair
- FGr Rooms
- Command units

• Unmanned circular sprinkler bar and water delivery systems:
- HFS units from BAR, LOS and LDS
- Circular sprinkler from LOS, EE and TF (with AB forest)

• MIK Crisis Management Coordination Center

**Special features**

The large forest fire in the wilderness area of the Brandenburg Nature Landscape Foundation has produced the following findings and impressions:

- Despite the moderate weather conditions for the summer months, the wildfire was able to spread to almost 100 hectares due to the concrete suspicion of explosive ordnance and burn deep into the moor and raw humus layer.

- The topographical terrain profile can also have significant effects on the dispersion behavior in the state of Brandenburg, despite the apparently small level differences.
  The terrain relief, characterized by moraines, influenced the local wind conditions.
  An unusual phenomenon was the emergence of a second fire front, which was still less intense than the downwind front, against the main wind direction.

- The mission tied up the predominantly volunteer forces (particularly from LDS) with varying numbers of participants for almost a month.

- In addition to the regular disaster relief units, modular units from LDS (e.g. Personnel module) as well as situationally assembled units from SPN regularly in the Ordered to deploy.

- Predominantly defensive measures were implemented to protect the unaffected wilderness areas. Part of the area was deliberately abandoned (balancing of the interests of protection) and left to burn.

- The use of HFS units and circular sprinklers has proven to be practical in this application and proved useful:

  - However, the Teerofensee (HFS water intake point) lost about 0.5 m of water. stand.

  - It is important to question whether persistent annual rainfall deficits will have a counterproductive, irreversible effect on surface waters used for large-volume firefighting water supplies.

  - Such favorable conditions (surface waters near the site of operation with sufficient access) cannot be expected across all forest areas in the state of Brandenburg.

- Firefighting support from the air was used to temporarily contain the Effective in burning litter and upper raw humus layer:

  - However, it was found that the follow-up extinguishing efforts from the air to extinguish the fire in the depths (ember pockets in the moor soil or deep humus layer) were not effective.

  - The coordination of aircraft by trained air coordinators was rated positively.

### 2.5. Forest fire near Falkenberg/Elster

A week-long series of fires in the area of the Rehfeld wind farm in the municipality of Falkenberg/Elster reached its peak with another forest fire report in the early afternoon hours of July 25, 2022. In addition to enormous area losses, the fire also spread to buildings.

#### Location description

In the early afternoon, a forest fire developed on a windward side of a forest area with a sparse to wide crown closure and a high level of calamity timber felling. The first-arriving firefighters from the Elbe-Elster district and the Nordsachsen district12 quickly identified a very rapid spread and immediately requested additional forces and resources. Attempts to contain the fire in areas with little vegetation failed due to increasing winds and increasing sun intensity. Just 45 minutes after the operation was initiated (1:28 p.m.), the fire department on site recommended

The municipal association's operations management notified the district of the declaration of a major disaster and requested additional disaster control units from other lower disaster control authorities. At this time, supra-local resources from the Elbe-Elster and North Saxony districts had already been deployed. Additional firefighting trains with tank fire engines from the regional control center area, assembled by the Lausitz control center, were dispatched in the afternoon. The uncontrollable forest fire continued to spread northeastward through the Rehfeld wind farm in the late afternoon. Due to the burning black areas absorbing sunlight, the overheating near the ground created its own thermals, drawing burning components across the light canopy. In addition, straw lying on harvested grain fields ignited and was set ablaze by the wind.

Red-hot and burning materials were carried more than 500 m downwind and deposited on combustible, thermally treated wasteland and grain land.
These "daughter fires" endangered the railway infrastructure (Falkenberg/Elster - Leipzig railway line) as well as the town of Rehfeld (including the individual properties along State Road 60 between Rehfeld and Kölsa). In the evening, around 7 p.m., a convergence line crossed the area. Rapidly increasing wind speeds of up to approximately 80 km/h and a change in wind direction from northeast to east caused extreme fire spread locally as a result of intensive radio surveillance and a wide, fully blazing band.

Apparently safe areas were destroyed by the fire over a distance of approximately 500 m
jumped over and ignited previously unaffected vegetation areas that were connected to structural
Facilities adjacent.

Due to the suppressed smoke (presumably downdrafts) and the ash and dust swirling over the entire scene, the emergency services in and around the forest had the opportunity to
A confusing picture. Because there was a real danger of being trapped by the fire, orders to withdraw were issued from the operational areas. The crew of a tank fire engine drove through an intense fire area in very poor visibility.

and managed to escape at the last moment. In this action, seven emergency personnel suffered
Smoke inhalation and minor burns. The emergency vehicle was significantly damaged by the heat.

At the same time, the evacuation of the districts of Rehfeld and Kölsa with Kölsa settlement was ordered.
The vegetation fire spread to a stable wing of a piglet breeding facility near Kölsa-Siedlung
and spread to other areas up to the property boundaries of single-family homes in the Kölsa settlement. The restructuring of the operational sections due to the withdrawal now had to
also take into account the building fires and hazards. The spread of the complex fire to other stable sections was successfully prevented, as was the fire

*12An inter-municipal firefighting assistance agreement for early mutual support already existed between the Verbandsgemeinde Liebenwerda (BB) and the municipalities of Beilrode and Arzberg (both SN).*

transition to residential development. A large number of livestock in the affected area died However, in the stable. At the same time, a stationary, A fully occupied passenger train was reported near the village of Rehfeld. The suddenly dying wind changed after about 15 minutes from east to southeast and favored a more moderate course in larger, poorly developed departments in the direction of Falkenberg-Lönnewitz airfield. These areas are designated as areas suspected of containing explosive ordnance. of the requested regional support units (especially BSE, TLF trains) in the evening and During the night, the other danger spots around Kölsa and the airfield were addressed. Final containment and securing of the fire perimeter areas was only possible In the following days, additional disaster control units, armoured clearance equipment and airborne support measures were deployed on the Brandenburg and Saxon sides, while fires were developing in inaccessible forest areas.

The affected forest area was cleared after completion of the operational measures on 2 August 2022 with 422 hectares were reported. Approximately another 400 hectares of heathland, grain, and fallow land burned.

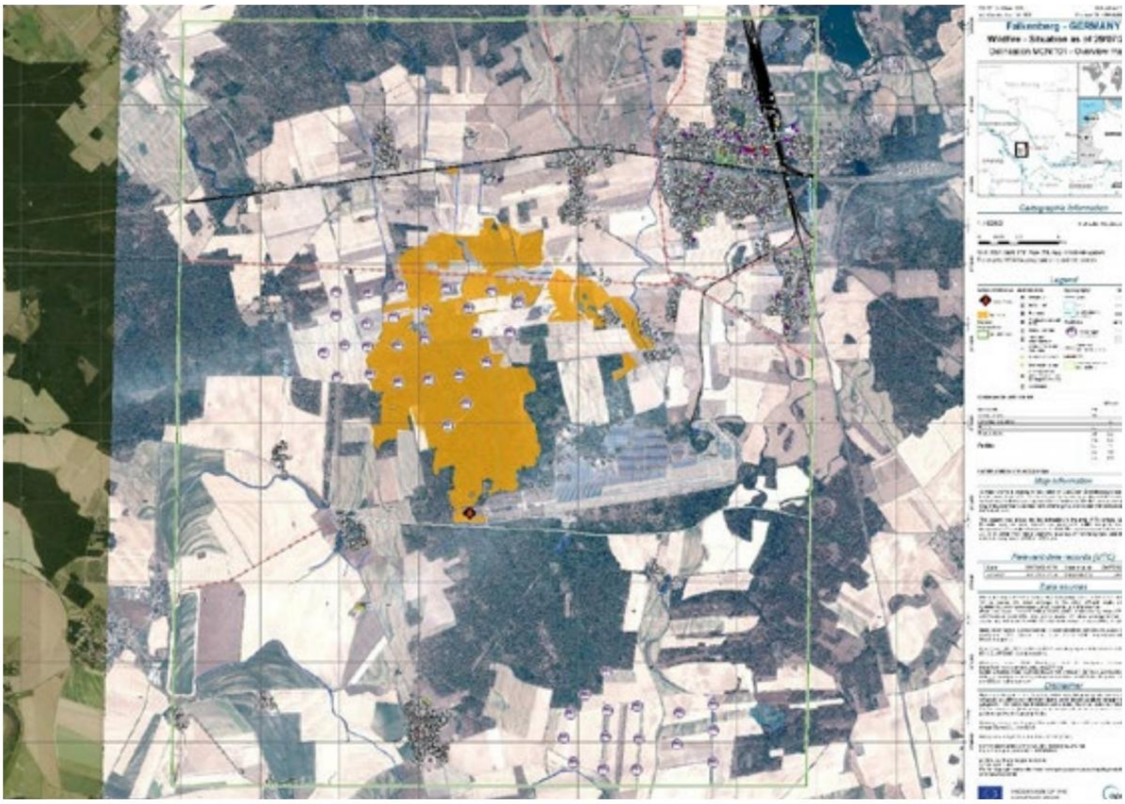


*Fig. 8: Fire area Rehfeld/ Kölsa 2022; Source: Copernicus Emergency Management Service (© 2022 European Union), EMSR611.*

**Forces and resources deployed:**

During the acute threat response in the dynamic operational phase, up to 550 emergency personnel were deployed simultaneously:

- Local fire brigades from the Elbe-Elster district
- Disaster protection units (BSE, leadership, SEG-V, SEG-water hazards) and situational compiled trains from other Brandenburg districts/independent cities:
  - Potsdam-Mittelmark district

- Oberhavel district
- Uckermark district
- Dahme-Spreewald district
- Oberspreewald-Lausitz district
- Spree-Neisse district

• Disaster relief units from the Free State of Saxony and the state of Saxony-Anhalt

• State police with:
  - Guard and shift duty
  - Police helicopter squadron for aerial situation assessment
  - Explosive ordnance disposal service

• LSTE and Brandenburg State Forestry Authority

• Emergency pastoral care from several districts

• Bundeswehr with:
  - District Liaison Command
  - Helicopter (including external fire water tanks) to support fire fighting combat and reconnaissance/coordination from the air
  - Coordination resources for helicopter deployment
  - armored clearing technology
  - Bundeswehr Fire Brigade

• THW with:
  - FGr Emergency Supply and Emergency Repair
  - Command units

• other external service providers and non-profit aid organizations:
  - agricultural and forestry contractors
  - @fire - International Disaster Relief Germany e. V.

• MIK Crisis Management Coordination Center

**Special features**

In connection with the operation at Falkenberg/Elster, the following knowledge was gathered and special measures were implemented:

• unfavourable weather-related influences with high outside temperatures (approx. 36°C), low relative humidity (approx. 27%) and partly gusty, rotating winds (gusts over 80 km/h) and three wind direction changes within four hours on the day of the eruption

• At times, running speeds of several hundred meters per minute were observed in the fire front (Note: this value corresponds to the scientifically proven

Rule of thumb13 v run= 0.1 xv $_{Wind}$ [m/s])

• Documentation of downdraft phenomena and extreme fire thermals (including axial rotation of the main smoke column during the passage of the convergence line)

• Destruction potential within a narrow time window and dimension of the destructive area (largest forest fire of the 2022 season)

• Risk of emergency services being trapped due to sudden changes in weather conditions

• Good fire water supply (fire water wells) distributed throughout the operational area through structural energy generation systems (as part of the approval process for wind turbines in the forest and PV systems at the airfield)

• Ad-hoc assembly of inter-municipal trains with tank fire engines (from several southern Brandenburg districts) by the Integrated Regional Control Center Lausitz to bridge the arrival times of the BSE (afternoon/evening of July 25, 2022)

• Suspicion/confirmation of explosive ordnance

• Use of circular sprinklers to create an unmanned barrier position along the Federal Highway B 183

• Evacuation order in the OT Rehfeld, Kölsa with Kölsa settlement

• Relocation of a Saxon command vehicle 1 (ELW 1) to the Technical Operations Management in Falkenberg/ Elster with the aim of direct exchange of information on situations and measures between the operational management in Brandenburg and Saxony (later also mutual deployment of liaison officers)

• Activation of the administrative staff of the Elbe-Elster district administration

• Controlled burning of the open land area on the former runway of the Falkenberg-Lönnewitz airfield (running parallel to the B 183) to secure the difficult to access forest areas south of the federal highway (creation of a safe control line)

13CRUZ, M.G., ALEXANDER, M.E. The 10% wind speed rule of thumb for estimating a wildfire's forward rate of spread in forests and shrublands. Annals of Forest Science 76, 44 (2019), page 2. https:// annforsci.biomedcentral. com/ articles/ 10.1007/ s13595-019-0829-8

# 3. Conclusions and optimization potential

In addition to the forest fires, which were quickly contained and their destructive potential suppressed, the large forest fires described above placed a particularly heavy burden on the integrated emergency response system for special hazard prevention in the state of Brandenburg. Above all, the structures, which are predominantly based on volunteers, formed the backbone of the response to dangers and damage to people, animals, the environment, and property. The operational management and ensured a coordinated implementation of measures appropriate to the circumstances. The proven cooperation and support of the Federal Police and the Bundeswehr, as well as the deployment of the THW in recent major disasters, is evident.

The 2022 forest fire season in the state of Brandenburg could reveal a predicted trend that could result in a more severe risk picture in this and the coming decades.
The years since 2018 have already generated an intense awareness of the threats posed by vegetation fires, which has led to organizational and technical developments14 in vegetation firefighting in the state of Brandenburg. In Brandenburg, evaluating the experiences from the forest fire years 2018 and 2019, a concentrated process of improving organizational and operational requirements for effective forest firefighting was initiated. Likewise, the MLUK has worked with the LFB
Numerous preventive forest fire protection measures have been implemented. The state has extensively supported both the measures for improved forest fire fighting and preventive forest fire protection measures. From the perspective of the state of Brandenburg,
a federal funding program for forest fire barriers to protect villages is needed,
to support municipalities and private owners in particular in implementing the measures mentioned. The federal government has also assisted in the preparation of a hazard and risk analysis on the potential hazard of Brandenburg's towns and municipalities for forest fires.
further assistance is offered, the initiative is welcomed by the State of Brandenburg
and the MIL will examine further steps in coordination with the federal government. The fire behavior of the Season 2022 in connection with the immediate threats to the population and
settlement areas has made it clear that further optimization steps will be necessary,
in order to adequately meet future challenges. In this context, interdisciplinary and inter-agency cooperation must be further intensified.

In order to prepare the structures and technical and organizational arrangements in the country for future To further optimize forest fire events and develop forest fire prevention based on current scientific findings, the creation of a Forest Fire Competence Center was proposed at the 2023 Forest Fire Summit, with the objective of:

- improving forest fire prevention,
- better planning of forest fire fighting operations and
- better coordination of all preventive forest fire protection and firefighting measures, as well as the optimization of the provision of central technical equipment.

In the mutual dependence of preventive forest fire protection and defensive forest

*14Preparation of a seasonal national plan for disaster control units and the current configuration of the Tank fire engine forest fire type Brandenburg*

For firefighting, coordination between the MIK and the MLUK is essential.

However, the departments and subordinate authorities and institutions dealing with forest fire prevention and fire fighting do not currently have the time and personnel resources to bring together the various fields of action in the required thematic depth and to revise them conceptually.

As far as the Forest Fire Competence Centre is provided with the necessary personnel, financial and The spatial resources to be provided could be pooled from the areas of the MLUK, the MIK, the DWD, the KMBD, the LSTE, and the area of science and research. The Forest Fire Competence Center could then facilitate institutional development and pooling of interdisciplinary forest fire expertise in the state of Brandenburg. For this purpose, the establishment of a Brandenburg Forest Fire Competence Center at the Wünsdorf location is a suitable option, as important expertise is already available locally with the LSTE's second school location, the State Forestry Authority, and the Central Police Service, including the KMBD, thus achieving significant synergy effects.

However, for such an implementation, it is necessary that the plans for the second school location of the LSTE are prioritized according to the urgent need and Appropriate funds must be allocated to the state budget in order to create appropriate conditions for such a forest fire competence center. Otherwise The competence center would simply take up the tasks of the previous forest fire working group and individual participants would be supplemented as needed.

The Forest Fire Competence Center could subsequently primarily conduct conceptual foundational work in preventive and defensive forest fire protection. In addition, relevant data and information could be prepared and accessed in a situation-oriented manner (e.g., digital situation maps and images, dashboard evaluations of past incidents, digital data analyses, etc.).

Based on the findings, the Forest Fire Competence Center could also provide support in training LSTE in vegetation fire fighting.

The Forest Fire Competence Center will be set up on an interdisciplinary basis, distinguishing between permanent full-time staff and temporary members. The concrete design of the Forest Fire Competence Center will require further coordination among the relevant departments.

### 3.1. Leadership organization/management

The command and IuK15 units (command and specialist service) of the lower disaster control authorities are performing significant work in organizing and structuring the operational measures and the operational area during the large-scale forest fires. In addition to their own units in their respective areas of responsibility, units from other lower disaster control units were also deployed at limited intervals in accordance with the state's seasonal planning.

*15Information and communication units (rapid response command support group according to the administrative regulations of the Ministry of the Interior and for Local Affairs for the implementation of the regulation on units and Disaster management facilities to the specialist management service from 16 December 2022)*

nung16 is providing support in the affected districts. The dynamics of the situation (partly also at
the days following the start of the operation) has placed great demands on these units. As can be seen from the example of As is clear from the almost parallel events in the Potsdam-Mittelmark district, several of these command units may be required simultaneously. The command and ICT units are designed for a 24-hour support mission (sometimes only 12 hours in the case of a
internal personnel changes in the supporting management unit) in order to relieve the personnel burden on the local management department.

**3.1.1. Situation analysis, information deficits and shift handovers**

During the daily shift changes between the units being released and the units being relieved, even with disciplined handovers, there is regularly a loss of mission-relevant information, which subsequently makes it difficult to trace the actions taken or to be taken.
In addition to the exclusive 24-hour deployment of this special personnel, the reason lies in the heterogeneous information processing and
Situational awareness is justified. In addition, the supporting command units and affected districts currently use various analog and digital situational awareness products. Ad hoc use and reuse of various software products by the

The necessary prior briefings and training, combined with the dynamic nature of the situation, could not be ensured by the respective replacement units, resulting in information losses. In addition, accessibility and procedures to and from mission-relevant
Stakeholders were not informed in a sustainable manner, so that replacing units were exposed to avoidable problem clarifications.17 Due to the constant changeover with new personnel after 12 to 24 hours, the management organization was extensively, but not always necessarily, restructured, as the initiating managers were not fully aware of the prior knowledge and reasons for the decisions.

Certain special maps (e.g., the LFB's forest deployment maps currently provided in the data cloud), which are relevant for situation analysis at various management levels, are difficult to research and, despite regular presentations, are partly unknown to the user group. The various geoportals of the state (e.g., Geoportal Forst Brandenburg, Geoportal Brandenburg) require extensive knowledge in their web application for the preparation of usable map sections and are partly limited by technical configurations.

The required scope of applications is limited and not intuitive to use. Furthermore, the web applications are not available in cases of weak or non-existent mobile network coverage.

**Solution 1:** A uniform management software used throughout the state of Brandenburg (currently being introduced: CommandX) will be used to optimise shift handovers
initiating and assuming management units (keyword: management personnel & ICT personnel according to seasonal state planning) by ensuring better traceability of decisions and measures taken through keyword searches and digital deployment logs. Furthermore, networking of various instances (local authorities, district, state) is possible and allows for role- and rights-dependent

*16-day planning of management, ICT and fire protection units among the districts for solidarity Support of forest fire-affected, lower disaster control authorities during the forest fire season for a 24-hour operation*

*17for example, in supply, catering, procurement, planned replacement of units*

*18for example in the LSTE seminar on vegetation fire fighting*

Insight into the developed command organization as well as an overview of the deployed units. Already documented operational information (e.g. section formation, deployed forces and resources with location and start of operation) can also be transferred seamlessly to a change in operational management from local to supra-local level.

**Solution 2:** Required data and map services must be easier to provide and use. Native and web-app-enabled map services for smartphones
can simplify intuitive use, but also increase acceptance, connectivity, and awareness. To defragment geodata information, the Geodata Infrastructure for Authorities and Organizations with Security Tasks in the State of Brandenburg (GDI-BOS BB) project must be further promoted across all jurisdictions as part of the introduction of a uniform management software.

**Proposed solution 3:** The current, exclusive 24-hour deployment period for command and ICT personnel must be further evaluated. To reduce information loss during shift handovers within an appropriate shift model19, it is necessary to examine whether the pool of command and command support personnel can or must be increased in principle beyond 24 hours in the operational area. However, it is the responsibility of the operational command to request additional replacement personnel at an early stage to ensure a seamless transition. This allows for the use of a consistent staff base and
minimizing training and information deficits in a rotating shift model
(appropriate deployment and rest times in the operational area). This increase in effectiveness and continuity could, for example, be achieved by two equivalent units working in shifts over a period of 48 hours (see also 3.2.1 Deployment and replacement times of the disaster relief units/ sustainability). As a minimal approach, only a core of personnel from selected functions could be deployed for a longer period (>24 hours).
(including rest periods) remain in the operational area. The Mobile Command Support Units (MoFüstE) to be established within the LSTE and within the regional control center areas could represent a central service complement.

### 3.1.2. KKM liaison staff (on-call system)

Since 2018, the LSTE has been providing liaison officers for the state government's Crisis Management Coordination Center (KKM) in the event of major disasters or catastrophes. This system has proven successful and serves as a means of exchanging information at short notice. between the operational command and the state as the highest disaster control authority. At the same time, the liaison officers specify and supplement requirements for forces and resources, and quickly clarify queries via short communication lines. Furthermore, the liaison officers support the requesting authorities in conveying capabilities.
and description. Furthermore, key leadership needs must be assessed jointly with the KKM. In some cases, the affected task authorities have explicitly requested and requested leadership support services from the LSTE.

*19depending on the frequency of high-stress phases and the resulting cognitive load on the forces (e.g. 2-shift model with 12h rotation or 3-shift model with 8h rotation or staggered individual function changes)*

*20Administrative regulation of the Ministry of the Interior and Local Government for the implementation of the regulation on Units and facilities of the civil protection (Civil Protection Ordinance - KatSV) for the specialist service Guided tour (VV-Fü) from 16 December 2022*

The availability of the LSTE liaison officers was confirmed at short notice via voluntary Assignments outside of regular working hours were ensured. During periods of high forest fire danger, concrete personnel planning for the LSTE in the form of an on-call system was not regularly mandated as a preparatory measure. Many initial requests were made after hours or on weekends, making recruiting at short notice challenging.

**Proposed solution:** In the course of the expected increase in support services of the LSTE in the event of vegetation fires (including the provision of special equipment such as 55 m³ Extinguishing water extraction tanks and command support) the introduction of an on-call system and its conditions (for example depending on the forest fire danger level during the forest fire season) should be examined.

### 3.2. Capability organization/management

The deployment of disaster control units to relieve the burden on local fire departments and to increase hazard prevention potential has become established in recent forest fire years. The seasonal pre-planning of disaster control units (specialized services management with ICT and fire protection), which has been carried out for years across all Brandenburg districts, offers those involved planning security and enables a high level of availability of personnel and equipment. At least for the 2022 forest fire season, it can be stated that the volunteer system of disaster control is being strengthened by the high frequency of deployments in
reached its limits in a short period of time from June to July 2022. In some cases, units from neighboring federal states were requested and deployed to the Brandenburg operational areas to provide support (e.g. in the town of Beelitz and the Verbandsgemeinde
Liebenwerda).

#### 3.2.1. Resilience of the disaster relief units

In the case of a pre-planned, 24-hour deployment of a disaster control unit in an operational area, arrival and departure times must also be added, so that an operation actually lasts 24 hours plus X hours (e.g. fire protection unit [BSE] UM travels to EE
ÿ 24 h operation + 6-8 hours total marching time). Within the 24-hour operation,
appropriate rest periods in suitable rest areas must be taken into account. In some cases, an internal personnel change takes place after a 12-hour deployment in a BSE, which
by the sending local authorities independently. However, there were also missions in which BSE and its personnel were deployed for 24 hours plus the arrival and departure time before
The duty of care of the deployed personnel must also be taken into account, which is why sufficient machine operators must be deployed. In particular, the machine operators must be given suitable rest opportunities and periods for regeneration before participating in public traffic in order to avoid microsleep and
to exclude similar things during journeys with the emergency vehicles.

In addition, after a 24-hour operation, the fittings used (especially distributors, jet pipes, but also suction and pressure hoses) are dismantled and taken away.
After arriving and being briefed on the area, the relieving BSE team reassembles all the fittings, which is a time-consuming process. Regular work during a 24-hour BSE deployment
will be reduced by three to five hours due to the daily setup and dismantling as well as the training.

gert. The overall effectiveness of the deployed units decreases. Similar findings could also be collected during shift changes of command units (see also 3.1.1 Situation analysis, information deficits and shift handovers). Congruent processes were also found in the catering units: The daily assembly and disassembly of nationwide The use of the same field cooking technology by the triggering and replacing rapid response group catering (SEG-V) proved to be a reduction in efficiency.

**Proposed solution:** Within the framework of the duty of care and efficient mission processing Alternative options and uniform specifications for the 24-hour to multi-day deployment of a disaster control unit (in particular, specialist services for management, fire protection, and support) must be identified. The 24-hour state planning remains in place until the event, independent of alternative options, but requires continuous review. The responsible operations command must derive a forecast of the expected total duration of the emergency response operation as early as possible within the framework of the situation assessment. If the forecast duration of the operation exceeds 24 hours, it must be examined whether shift-capable reinforcement (additional,

simultaneous BSE in the operational area) or an internal BSE personnel exchange (extension of the Deployment time beyond 24 hours) can be carried out in order to ensure the 24-hour deployment of a unit and to transfer it to a shift model of operational and rest periods in the operational area. A suitable shift model must be selected between the units on site,
This takes into account the physical strain on the emergency services, the logistical effort required for the changeover, and the time of the changeover (changeover at night is unsuitable). The KKM should be informed of this in order to facilitate possible coordination21 between all participating lower civil protection authorities (e.g., through video conferences).

Within the framework of the planned shift models, it is also necessary to check to what extent the material and technology used initially remain at the site and are passed on to the following shift can be handed over. In this respect, subsequent return and, if necessary, repair must be ensured.

### 3.2.2. Inter-municipal TLF trains (TLF-Sofort)

During this forest fire season, large forest fires repeatedly resulted in ad hoc alerts for water-carrying vehicles (tank fire engines - TLF) to provide mobile firefighting water supplies in conjunction with direct firefighting. TLF units from several local authorities and other agencies (in this case, the Federal Fire Service) were assembled as needed to initiate concentrated firefighting measures along the fire perimeter. This combined unit resembled

basically the first sign of a defined BSE, but these represented a special Ability for immediate intervention. The units came from both the affected district as well as from neighboring districts and independent cities to determine the marching times and routes and therefore did not directly correspond to the deployment preamble of a BSE. However, these TLF platoons bridged the marching times of the BSE during critical operational phases and successfully supplemented the local units while maintaining the local basic protection.

**Proposed solution:** It is necessary to evaluate whether a separate TLF train should be provided for each district, similar to the

*21This could include increasing the sustainability of a disaster management unit beyond the practiced 24-hour operational area.*

TLF trains of the professional fire brigades in the country (including command vehicle) for immediate support can be provided in order to implement timely response measures (approx. 2 hours after alarm) in the adjacent or own area of responsibility to avert danger in the event of forest fires without undermining basic protection in one's own or neighboring local authority. The deployment time of such a TLF train, which is to be defined, should not exceed 12 hours due to the ad hoc alert. Within this time frame, an organized follow-up (personnel inquiries, compilation of catering, supply and accommodation components, marching times) of required BSE to be expected. Similar processes exist in pre-hospital emergency rescue in the event of a mass casualty incident (MCI) with supra-local MCI emergency aid.

#### 3.2.3. Development of capability modules for fire protection units

The fire protection units (BSE) of the Fire Protection Department have provided valuable personnel and technical support to the affected local authorities during the last major forest fires since 2018. The disaster control unit concentrates mobile and semi-mobile firefighting capabilities using water-carrying tank fire engines (a total of approximately 24,000 liters of extinguishing water), firefighting or defense via stationary firefighting vehicles, as well as water supply over long distances (a total of approximately 3,200 m of B-pressure hose) in the case of vegetation fires. Additional command vehicles complement the unit, which comprises at least 73 emergency personnel. As a result of the events since 2018, some fire service and disaster control leaders considered the capabilities, which are organized into platoons, to be too rigid and not integrable in every situation.
As a result, various lower disaster control units supplemented the BSE with personnel and technology. A heterogeneous BSE structure emerged in the country, which differed from the minimum the administrative regulation of the Ministry of the Interior and for Local Affairs for the implementation of the Ordinance on the Units and Facilities of Civil Protection for the Fire Protection Service (VV-BS) of 16 December 2022. The disadvantage here is the differentiated overall strength of the unit, which, in the case of shift replacements and the simultaneous lack of transmission of the district-specific lists of planning challenges (e.g. equal section allocation, supply/catering). It proved advantageous
In most cases, the greater overall tactical operational value of the additional units. By forming modules of the same size, the lower disaster control authorities were able to react flexibly to the operational situation and local circumstances and also install timely self-sufficiency capabilities. For example, in the Dahme-Spreewald district, a module
Emergency personnel were called in to manually set up a kilometer-long, unmanned circular sprinkler system.

**Proposed solution:** The VV-Kats explicitly allows an extension of the BSE, so that the Description of operational modules as special capability additions is permissible.
In order to maintain the uniformity and comparability of tactical capabilities of the BSE throughout the country, it should be examined whether the additions can be described in a concept catalogue and can therefore be adapted by the lower disaster control authorities. The idea of greater flexibility could thus be met, since the BSE
In addition to vegetation fire operations, they are intended to be used in other major disasters/ catastrophes such as floods and train accidents. Examples include the module developments in the Dahme-Spreewald district. The BSE will be divided into

integrated into the modules of readiness management (two-stage), fire fighting (LF component in each case, TLF component), fire water supply, technical train, emergency services and logistics.

### 3.3. Airborne support

Aerial support for reconnaissance and firefighting purposes in forest fires has increased significantly and gained importance in recent years. Depending on the operational circumstances, this can provide valuable information for assessing the situation at an early stage, and from this, effective measures can be taken to reduce the spread of fire in difficult-to-access or particularly vulnerable areas. Currently, helicopters of the BOS (Brandenburg State Police, Federal Police, and Bundeswehr) are being used in particular. Drones procured by municipalities or provided by aid organizations have supplemented aerial reconnaissance measures in the past. Above all, aerial firefighting measures can only be understood as support for ground-based measures. The potential for explosive ordnance to be released as a result of heat exposure must be considered three-dimensional and pose a safety threat to aircraft crews.

The increased frequency of use of manned and unmanned aircraft in recent years has resulted in the development of the function of the ***air coordinator*** in Brandenburg. For this purpose, a special course is offered at the LSTE "Technical Advisor and Air Coordinator
at fire and disaster relief sites." The following training content, among others, must be learned within 5 days: legal basis, command organization, accident prevention, helicopter deployment by the police and armed forces - meteorology, map reading, radio - plan exercises, practical instruction. This function serves the operations command as a technical advisor and the air section management in order to evaluate the special measures and requirements of air support and to initiate them on behalf of the mission. However, it should be noted that the short-term availability of manned aircraft cannot be guaranteed at all times or can be delayed in some cases due to the statutory primary mission of the dispatching BOS.
after the highly dynamic operational phases. To make matters worse, air support was occasionally requested at KKM in the late afternoon and evening hours, so that with the onset of darkness, only a short operational duration of the rotary-wing aircraft
Limited decision-making powers and skepticism about cost coverage may be considered as possible causes.

#### 3.3.1. Aerial reconnaissance

For airborne, qualitative reconnaissance purposes, rotorcraft from the helicopter squadron of the

The Brandenburg State Police deployed the helicopter. Furthermore, the helicopter allows for the transport of an additional observer. Previously used helicopter and drone images tie up the few specialists experienced in technical analysis, requiring considerable time and delays to manually extract essential information and operational priorities from these data sets. This season, airborne coordination of the measures was carried out for the first time with a Bundeswehr SAR helicopter during the

A major forest fire in Falkenberg/Elster was successfully tested. This approach increased the efficiency of extinguishing water drops. However, the use of lifting

The use of drones for this task would be associated with considerable costs. Consideration should therefore be given to integrating drones even more extensively for such reconnaissance and coordination tasks in the future. However, the joint use of helicopters and drones requires considerable coordination and coordination.

**Proposed solution:** To conserve resources and ensure the availability of this special technology for For the primary, statutory purpose, drone concepts ready for implementation have already been developed for future reconnaissance purposes (mapping of burnt areas, perimeter determination including evaluation of the risk of spread at the fire edges, hotspot search) within of the State of Brandenburg, so that a funding of one equipment vehicle drone for each lower civil protection authority with a flat-rate subsidy of 70% of the costs in 2025 and 2026 will be provided. Further issues within the State of Brandenburg

should be evaluated. One example is OPTSAL22 of the German Aerospace Center (DLR) with its project partner EUROCOMMAND. The use of satellite images, for example, via the European Union's Copernicus Earth monitoring program, is useful, but may be subject to a delay in availability (request via the GMLZ; the next possible, cloud-free overflight may be the next day).

In this respect, new technological developments for airborne reconnaissance must be included in the overall assessment, which offer real added value and an increase in effectiveness for Indication of operational priorities (software-based evaluation of aerial photographs to determine the situational vegetation and its condition [residual moisture content, crop density, condition/seasonal cultivation on agricultural land]) and thus for the tactical concentration of units. In addition, in special operational situations

to consider the possibilities of air coordination from a light multi-purpose helicopter with thermal imaging sensors in order to improve airdrops.

#### 3.3.2. Request procedure for air-based measures

The requirements regime of the flying BOS deployed so far in the state of Brandenburg During the initial request, some were perceived as too lengthy. Early air support can significantly reduce damage.

contribute when ground-based units are considered insufficient. The deployment of aircraft for firefighting from the air after the day of the fire outbreak should be critically questioned in some situations. Rather, the operational focus must be

early integration and availability of aircraft.

**Proposed solution:** The communication and procedural processes of the currently used request channels must be reviewed for optimization potential. At the very least, the request form should be evaluated with a view to making it more intuitive to fill out with fewer free text components.

#### 3.3.3. Determination of drop zones

Although the coordination of helicopters for firefighting support by the The function of air coordinator in the state of Brandenburg has been significantly upgraded, further management functions are required for clear coordination between the ground-based

22 https://www.optsal.de/

The air coordinator and the aircraft crews are required. Clear communication of the situational drop target areas and the effectiveness of the extinguishing water drops can only be achieved with difficulty using coordinates for both the air coordinator and the aircraft crew. Furthermore, for their own safety, the aircraft crews are
rely on current maps of suspected explosive ordnance sites.

**Proposed solution:** For the transmission and coordination of aerial water drops, suitable maps (e.g., forest fire maps with areas suspected of containing explosive ordnance) should be provided with a dense grid of quadrants. An example of this is the open-source situation visualization solution23 with a grid layer.
It is necessary to examine the possibilities for integrating such a tool. The CommandX management software and the geodata infrastructure for BOS (GDI-BOS BB) are particularly in focus. Furthermore, options with the Brandenburg State Surveying and Geoinformation Agency (LGB) and the Brandenburg State Forestry Agency (LFB) are being evaluated.

### 3.3.4 Involvement of service providers

During the highly dynamic operational phases, the ad hoc availability of manned aircraft from the emergency services was not always possible due to the statutory primary mandate of the dispatching emergency services, or in some cases, it was only possible with a delay. Due to the expected increase in extreme fire behavior during vegetation fires in the medium term, early intervention by ground and/or air units can lead to operational success, as future fire factors are likely to make the fire uncontrollable just a few hours after it breaks out.

**Proposed solution:** As an alternative to the flying BOS units, the contractual
The involvement of service providers should be examined. As part of permanent availability, rotary-wing aircraft from private providers could be used during the forest fire season (if necessary).
The possibilities for using such service providers should also be examined, especially by districts at risk of forest fires. This should also include consideration of the integration and responsibilities during operations (liability issues, position within the command organization, authority to issue instructions, technical and organizational communication requirements).

However, this entails considerable costs in terms of maintaining the reserve, which municipalities are generally unable to cover. At least in Brandenburg, the state lacks the necessary authority to allocate operational resources for aerial firefighting below the threshold of a disaster. Therefore, the above-mentioned proposal should be examined to determine whether and to what extent additional aircraft models for vegetation firefighting can be procured jointly with the federal government. The procurement of five additional helicopters for the Federal Police under

Proportional cost sharing by the states could increase the availability of appropriate operational resources within the Federal Police. The states' cost sharing should not result in additional operational costs, which would shorten decision-making processes and, if necessary, allow for a low-threshold deployment to be initiated at an early stage. However, this would require broad consensus between the federal and state governments.

*23 http:// lavisu.de/*

#### 3.4. Explosive ordnance (suspected)

In retrospect, almost all large forest fires in the intensive forest fire years since 2018 affected areas suspected of containing explosive ordnance.
In addition to the fire regime expected in the future, this represents one of the greatest challenges of the coming years and decades. A comprehensive freedom from explosive ordnance can be
The state of Brandenburg cannot be implemented in the short term. This remains a task for generations. Direct firefighting in areas suspected of containing explosive ordnance is difficult due to the safety distances that must be maintained. When there is a latent risk of heat-based explosive ordnance being deployed, both direct ground-based and airborne measures are limited. The explosive effect is assumed to be three-dimensional, so that airborne support for containing fire areas is also subject to corresponding safety regulations. Airborne measures do not usually extinguish a forest fire, but merely serve as a technical and tactical supplement to ground-based measures. Follow-up extinguishing operations by rotary-wing aircraft are considered to be less economical overall, in addition to the aforementioned safety aspects.

The cooperation between the Explosive Ordnance Disposal Service of the Brandenburg State Police Central Service (KMBD), authorities and organizations of non-police emergency response, as well as with other specialist authorities, has intensified considerably and steadily improved since 2018. The KMBD provides expert advice on, as well as the recovery and storage of found ammunition, in a timely and proportionate manner. The KMBD's recommended safety distances are selected on a case-by-case basis based on existing knowledge of former military land use and ammunition use, as well as the risk situation of the vegetation fire, and on the basis of the relevant regulations. The cooperative interaction of the KMBD and the authorities involved in vegetation fires contributed to the success of the operation, while observing the principle of proportionality and the safety of emergency personnel. In individual cases, the KMBD accompanied or initiated road surveys and enabled emergency personnel to

This led to faster operational success.

### 3.4.1. Early involvement of the KMBD specialist advisory service

The operational commands regularly made use of the KMBD's expert advice in the case of vegetation fires on areas suspected of containing explosive ordnance. The KMBD's request was
Not always promptly, but sometimes with a delay without an adequate description of capabilities. The KMBD has not yet introduced an on-call system for vegetation fire incidents, so that only through internal coordination (even outside of normal business hours) could available and authorized personnel be dispatched at a later time, while ensuring the statutory primary mission. In some cases, the requested advisory orders from the operational command could not be implemented due to the KMBD's human and technical resources, or could only be implemented through a statewide concentration of personnel at the scene. Furthermore, arriving KMBD personnel were not regularly provided with a technical briefing.
with the intended mission objectives, so that delays in the situation assessment and ultimately in the support services had to be accepted. The advice and support of the KMBD ensures an efficient, but at the same time safe
Firefighting on vegetation areas represents an important basis for decision-making and implementation, which should be used as early as possible.

**Proposed solution:** The KMBD should examine whether a seasonal on-call system for vegetation fires can be established (if necessary, depending on the forest fire danger level; see 3.1.2 KKM Liaison Personnel). The resulting reduction in dispatch times is necessary for the earliest possible consultation and support in order to assess further needs. To this end, the operations management must ensure regular and qualified involvement of the KMBD, for example, through checklists.

### 3.4.2. Data exchange of suspected explosive ordnance sites

The LFB maintains current, information-based data sets on suspected explosive ordnance sites into the forest fire protection maps and has made them available via a restricted-access link since 2022. Nevertheless, the local use and storage of this data is very heterogeneous.

Dynamic maps (regular evaluation of entries and exits) of suspected explosive ordnance areas are available to the local fire brigade units (first arriving units at a vegetation fire) is not available across the board in all local authorities.
This complicates the initial assessment of the risk to the individual and the necessary initial measures.
On the one hand, supra-local and even local authorities sometimes fail to forward the maps and data sets provided by the KMBD to the local fire departments. On the other hand, the temporarily valid map contents must be treated as sensitive data sets that cannot be made publicly available.

During two extensive vegetation fires on areas contaminated with explosive ordnance near the Saxon-Brandenburg state border (Verbandsgemeinde Liebenwerda), Brandenburg emergency services were able to access maps from the Brandenburg KMBD. Maps of the Saxon explosive ordnance areas across the state border were only available available with a very long delay.

**Proposed Solution 1:** Local fire departments must be provided with easy access to current map and data services, at least within their areas of responsibility. Even though the availability and exchange of updated explosive ordnance maps can be organizationally challenging, alternative sources of information must be identified. One possibility could be the LGB datasets via the currently developed

Geodata infrastructure for authorities and organizations with security tasks (GDI-BOS BB) Suitable data sets can be imported into the situation display module of CommandX (EUROCOMMAND) and made available only to authorized users.
There is also the possibility of simplifying the update service. The KMBD and the LFB should be involved in this process.

**Proposed solution 2:** Intensifying cooperation between the Saxon and Brandenburg KMBD and the Brandenburg fire departments in the border region is desirable. Individual measures should be developed in a goal-oriented manner among the parties involved.

### 3.4.3. Efficient, safe tactics/special techniques

Direct firefighting on areas suspected of containing explosive ordnance is difficult due to the safety distances that must be maintained. In the case of a latent risk of heat-induced explosive ordnance being deployed, both direct ground-based and air-based measures are required.

limited. The explosion effect is assumed to be three-dimensional, so that airborne support for fire containment is also subject to appropriate safety regulations. Airborne measures do not usually extinguish a forest fire, but rather serve merely as a technical-tactical supplement to ground-based measures. In addition to the aforementioned safety aspects, follow-up extinguishing operations by rotary-wing aircraft are considered less economical overall.

**Proposed solution:** Technological alternatives to direct firefighting on UXO-contaminated areas must be examined. To rule out personal injury, the evaluation should focus particularly on unmanned systems. In addition to remotely operated tracked chassis24 (possibly with extinguishing water tanks for firefighting with extinguishing water and mulcher attachments for wound stripping), the operational tactical values of firefighting drone swarms25 (current research project in the state of Brandenburg) must be investigated. Furthermore, the procurement of manned, armored vehicles with additional configurations (extinguishing water tanks, attachments, etc.) for reconnaissance and firefighting purposes must be assessed. Such systems can be used to reduce the risk of personal injury.

for extinguishing effectiveness directly at the source of the fire and for rapid operational success
contribute and current procedures (cost-intensive and risky rotorcraft use)
Complementing the concept, manned, armored reconnaissance and firefighting vehicles (configured forwarders26 with fire water tanks and monitors, tank firefighting vehicles27) could be considered for controlling and supplying the unmanned equipment. The use of unmanned hose-connected tracked chassis should be critically examined due to their limited maneuverability and the need for a stable water supply.

The KMBD should be included in the evaluation process due to its technical expertise. If the procurement vote is positive, the establishment of a state-owned special unit at the LSTE and/or the KMBD should be evaluated. It should be examined whether the technology to be procured can also be incorporated into the fire protection concept of the KMBD's ammunition dismantling facility in Kummersdorf-Gut. The states, as well as local and supra-local authorities bordering the states, should be mutually informed about the capabilities of their neighboring units. For the state of Brandenburg, the publication of a brochure is planned within the framework of the recently amended Disaster Management Ordinance with the respective service-specific administrative regulations. This brochure could

also be distributed to neighboring federal states. If necessary, the state fire brigade associations could assist with further distribution. Publicity events
Events such as the planned FIREmobil trade fair in Welzow or FLORIAN in Dresden could represent suitable additions to expand knowledge. Each federal state has structured and developed special capabilities, including vegetation firefighting, within the framework of disaster control. The knowledge of the capabilities of the respective disaster control units of a neighboring federal state is currently still limited; the establishment of the GeKoB will bring about improvements in this area.

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*24 https:// dok-ing.hr/ defence-security/*
*25 https:// www.naturetec.flights/*
*26 https:// www.youtube.com/ watch?v=Z2vfU2zM-p0*
*27 https:// www.czdefence.com/ article/ czs-15-triton-introducing-new-reinforced-czech-water-tenders*

### 3.5. Training/Tactics

Since 2019, the LSTE has been offering a special seminar on vegetation fire fighting, which is continuously being developed in order to raise the necessary awareness in the local fire brigades and to train multipliers for basic training on site. In addition, the LSTE offered a training program in air coordination for the first time in 2020. The aim of the training is to train managers of public fire brigades and other Disaster relief organizations involved in providing expert advice and management of airborne hazard/damage prevention measures (particularly in the case of vegetation fires)
In the 2022 training year, a further, more advanced course was offered. This course already incorporated insights and impressions from the extensive use of aircraft during this year's forest fire season, which will be carried out in a joint evaluation event at a direct working level with the previously deployed flying BOS (Federal Police, Police of the State of Brandenburg, Bundeswehr) and drone experts in the State of Brandenburg.

Vegetation fires pose particular challenges due to their dynamic nature and progressive spread. However, firefighters have so far been trained and conditioned primarily for static situations (e.g., building fires, traffic accidents). Recent events have demonstrated the need for a further intensified training campaign for vegetation firefighting in order to convey, on the one hand, safer procedures and, on the other hand, efficient response (tactical toolbox for vegetation fires). At the same time, tactics from abroad must also be considered in relation to local migration efforts.

#### 3.5.1. Development of necessary training products (context: vegetation fires)

Within the framework of the training concept, the LSTE is actively involved in the sub-working group Training/Tactics (UAG) within the open-country working group National Forest fire protection and provides input, supported by UAG members, for a nationwide standardized training and teaching approach in the newly evaluated field of vegetation firefighting. Working cooperatively, the UAG evaluates tactical alternatives for implementation in local regions. The stated goal of this UAG is the development of nationwide curricula (supplements) and recommendations for existing and newly developed specialized training courses.

**Proposed solution:** If the implementation progress of the UAG does not match the LSTE the desired dynamics, the LSTE aims to achieve this, probably in the training year 2024 to be able to offer special training courses. The focus will be on the courses "District Instructor for Vegetation Fires" and "Leading in Vegetation Fire Operations." The District Instructor Course is intended to establish basic training at the local and supra-local level, which will contribute to increased risk and safety awareness and The course is intended to convey the handling of situation-adapted, technical-tactical implementation variants. The handling of centrally managed firefighting technology (see 3.6 Central State Procurement and Technology). The training objective of the leadership course is to acquire skills adapted to the leadership levels in selecting tactical variants. In addition, the leaders must be sensitized to the situation assessment of these dynamic operational situations and also incorporate extensive weather influences into the decision-making process.

decision-making. The technical-tactical operational values of centrally procured
Special technology must be understood and used by managers in a way that is appropriate to the circumstances can be.

### 3.5.2. Tactical fire

Given the extreme fire behavior observed during this forest fire season (e.g., Treuenbrietzen, Beelitz, and two fires in the Liebenwerda municipality), in addition to the technical-tactical options already known and established in Germany, previously unknown options in these latitudes must be evaluated for implementation in the state of Brandenburg. This includes, in particular, tactical fire deployment as a controlled burning

substance reduction during an extensive vegetation fire to directly prevent the spread in the main direction of fire propagation or to secure defined areas.
The effectiveness of this option is regularly demonstrated, particularly in European and Anglo-American countries. Even if not every vegetation area in Brandenburg is suitable for
While tactical fire control (especially controlled burnout and tactical pre-fire control) is suitable, this expands the options for action, particularly in weedy, sparse pine forests. During the forest fire in the town of Beelitz, tactical pre-fire control represented the last and most successful option[28] to prevent the fire from spreading to settlement structures. At the same time, this measure proved to be very efficient in the use of resources, taking into account the length of the fire front area.

Forest fire in the Verbandsgemeinde Liebenwerda (local municipality of Falkenberg/
Elster) an open area of approximately 15 hectares was created to prevent fire from spreading to
A large, contiguous pine forest was burned in a controlled manner to reduce the fire load. Given the predicted, prolonged drought periods and the resulting reduction in firefighting water availability, tactical pre-fire operations may represent a future option.

**Proposed solution:** Since tactical fire deployment currently represents a secondary technical-tactical option, centrally available LSTE employees should initially be qualified for this application. At the same time, however, it can be assumed that this option will assume greater importance by the end of this decade. However, this option requires a lengthy development of skills and experience, which can currently be achieved primarily in southern European countries with forest fire experience. In preparation for the expected increase in forest fire risk, the qualification process should begin promptly. Meanwhile, an accreditation procedure for the recognition and qualification standardization of such skills must be developed in the Federal Republic of Germany. It is important to prevent emergency personnel from being qualified by companies and organizations without a

Comparability and minimum standardization are essential. Due to the inferior training, supposedly qualified personnel could underestimate the complex application of tactics and cause serious dangers. Tactical fire operations are also a common option for targeted, preventative fuel reduction.

to evaluate constructively and impartially with the relevant ministry and the State Forestry Authority.

*28implemented by @fire*

### 3.5.3. Development of tactical action standards

For vegetation firefighting, universal to specialized fire engines with crews of three to nine people are currently used as one of several units. Tank fire engines for forest fires (e.g., TLF-W BB) in particular offer space for three people. The presentation of a rapid fire attack with pressure hoses over larger

Distances from the vehicle location (ÿ 100 m) are difficult to achieve with this extended squad. For this purpose, additional vehicles with a squadron or group strength (some with road chassis) until at least one group equivalent value (0/1/8/9) is established.
A standardized task allocation for this vehicle composition and type of operation takes place
For classic, static fire operations and traffic accidents, standards are set in accordance with Fire Service Regulation 329 or the vfdb Guideline 06/0130
For dynamically spreading wildfires, comparable conventions are currently lacking, which would facilitate the actions of the crews and the responsible commander for a regular vegetation fire and increase the safety of the emergency services.

**Proposed solution:** For the state of Brandenburg, standards of action should be established for units consisting of one or more vehicles. These regulations, which apply equally to training, exercises, and operations, should also include basic specifications for predefined attack and retreat routes, safety zones, and secure posts in order to the protection of the emergency services at all times and especially in the event of changes in the situation (areas with high fire intensity, changing wind direction/fire progression).

### 3.5.4. Participation of specialized non-profit organizations

The fire behavior during this forest fire season has shown that existing resources can quickly reach their limits if current tactics are maintained. The dynamics of the local fires were comparable to extreme vegetation fires in southern Europe and the Anglo-American region. There are currently only a few specialists in Germany who can draw on such experience. Some

Non-profit organizations have already gained extensive experience in vegetation fire fighting over the past decade and have specialized through training according to foreign standards31 and internships in Europe and the USA. Particularly in Brandenburg the non-profit organizations @fire - International Disaster Relief Germany eV and the Waldbrandteam eV have concentrated extensive expertise in this complex.
These organizations have already demonstrated their capabilities this forest fire season through several deployments to large forest fires. Knowledge transfer with Brandenburg emergency services on efficient firefighting has already taken place during the deployment.

**Proposed solution:** Non-profit organizations should be integrated into the central vegetation firefighting training and specialized vegetation fire response, depending on their capacity and willingness to cooperate. Selected training content for the LSTE courses to be developed (see 3.5.1 Development

of necessary course products) could be provided by their qualified members
In addition, the establishment of a central state management unit for vegetation fire prevention

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*29 units in firefighting and rescue operations*
*30Technical - medical rescue after traffic accidents*
*31insbesondere NWCG (National Wildfire Coordinating Group) der USA*

The organizational connection could be made at the LSTE. In addition to fire analysis, fire perimeter mapping,
Tactical fire support, air coordination, and specialized guidance during post-fire operations are part of this unit's capability portfolio. However, the medium- to long-term capabilities of these organizations must also be regularly evaluated, as members are organized from throughout Germany and large-scale simultaneous operations are expected in the future.

### 3.5.5. Training opportunities for staff work and operational section management

The training of functionaries at the various management levels, which develop from the growing management organization, is solid and complies with the requirements of Fire Service Regulation 2 (Training of Volunteer Fire Departments), which has been introduced as mandatory in the state of Brandenburg. Nevertheless, individual deficiencies within the staff line organization were evident in the predominantly volunteer positions during the major forest fire incidents. In addition to routine and consolidated functional performance in a technical operations command or within a command staff, the analysis must also focus on the functions of a section management as an intermediate authority within the management organization. The managers must develop a sensitizing understanding of their role for major disasters and

The LSTE has already overcome this deficit after the
The forest fire years 2018 and 2019 were identified and a specialist publication32 was published in spring 2021 to improve the interpretation of operational command with command units at command levels B and C in accordance with Fire Service Regulation 100. Provided and adaptable templates for better and clearer task performance (e.g. supply notification, written command scheme) within the respective command organization complete the reading. Since the
The publication is experiencing increasing popularity. However, it is clear that
that these templates are not yet being used across the board.
In order to ensure analog redundancy in the event of a complete failure of digital situation display and documentation options, it is desirable to use them as uniformly as possible.

Furthermore, based on the experience gained in 2022, it is evident that the proportion of predominantly volunteer officials in the specialist management services of the lower civil protection authorities appears to be stagnating or even declining slightly.
This may be due to the fact that many of the volunteer leaders, who are primarily involved in public fire services, already perform technical and administrative work and may find the dual functions resulting from membership in the specialist management services a burden. In addition, the lower

Disaster management authorities should generally be required to offer regular training opportunities within their organizational structures in order to consolidate procedures and functional performance through exercises and training.

**Proposed solution:** The LSTE specialist publication (including templates) should be further integrated into training and continuing education programs and promoted to the respective task forces. In addition to the pure training courses, the LSTE should also develop continuing education programs for specific functions in the staff departments and subordinate command units.

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*32 https:// lste.brandenburg.de/ lste/ de/ service/ fachinformationen/ fuehrungsunterstuetzung/ #)*

in order to develop the role competencies of the respective managers more strongly. The integration of adapted, functionally appropriate training content for the use of the CommandX management software must be considered. Increased staff training by the LSTE and the BABZ (formerly AKNZ) can help the responsible authorities and the districts to better handle operational situations. The cross-state (staff framework) exercises of the operational commands and the designation of joint locations for both operational commands in anticipation of operational scenarios should be promoted. The intermediate or preliminary stage of the bilateral exchange of liaison officers (with increasing personnel numbers) will be implemented and planned in advance. The creation of a training course "District Instructor in Vegetation Fire Fighting" at the LSTE, involving @Fire and other specialists (SKV, etc.), is also planned.

### 3.5.6. Handling digital radio

Voice and data services via BOS digital radio are one of the most important means of communication for police and non-police emergency response agencies. Digital radio terminals are available nationwide in vehicles and command centers. The technical communications organization can be adapted to the command organization at the scene of the incident, from the initial local deployment phase to the potential escalation into a major incident/disaster. For this purpose, tactical requirements of the BOS digital radio users can be presented to the authorized agencies prior to a thorough operational preparation. The Authorized Digital Radio Agency (ASBB) is available to the local and supra-local authorities in the state of Brandenburg as a reliable partner. In particular, the radio technology requirements along state borders require intensive, equal coordination with the respective authorities in the neighboring federal state. The ASBB has provided the neighbouring federal states with call groups from Brandenburg, release non-police fleet mapping for transfer and use in their structures. Neighbouring federal states are free to program these call groups on their terminals. Currently, there is no clear overview in the state of Brandenburg as to which neighbouring federal state has programmed the released call groups from the state of Brandenburg into its specific Fleetmapping has taken over. In order to determine the technical possibilities and troubleshooting in the case of use, such an overview can increase the efficiency of the technical This can contribute to communication between task forces along a national border in a cross-border emergency situation. Furthermore, the ASBB already enables event-specific group combining for specific emergency scenarios. In addition, dynamic call group assignment can be implemented in individual cases. These technical options are available nationwide and can be tailored to tactical requirements.

**Proposed solution:** The ASBB and the LSTE should provide further training materials for individual self-study. This includes freely available tutorials that explain handling and simple troubleshooting through short videos. For example, tactical call groups for operational-tactical cooperation (TBZ call groups) or special cooperation (ZA call groups) could be reserved or preconfigured for cross-border operational situations. This need is discussed with the affected

These technical optimizations allow for regional, mission-specific preparations (e.g. switching call groups when alerting to

deployment sites near the border), which should be taken into account in coordinated special plans.
Specifically, the following measures can be implemented in the short term:

- Revision of planning documents (including cross-border ones) (hazard prevention planning, communications plan). This includes taking into account the prepared radio communication measures, such as:

  - to project the "ZA Group" (Cross-border Digital Radio Group) onto the terminal devices program and have them released by the ASBB,
  - Involvement of the ASBB in the communication planning of the districts,
  - Approval and designation of "TBZ groups" (tactical-operational cooperation groups) by the ASBB and subsequent integration into the communication plan,
  - In-depth training of the fire brigade in the operation of radio technology,
  - In-depth training at the state fire brigade schools for staff members and Communication planners at the district headquarters and operations management,
  - In individual cases, the ASBB can compensate for limited radio coverage an increase in transmission power can be requested,
  - Mobile TETRA base stations (mBS) can be deployed to improve local radio coverage if suitable locations are available. A federal satellite-based mBS is stationed in BB and can be deployed if available. Additional state-owned systems are located in the

    Procurement.

**3.5.7 Participation in European knowledge transfer**

A continuous transfer of knowledge/exchange of experts with foreign specialists and special units can make it easier for the State of Brandenburg to avoid mistakes in the implementation/introduction of new measures or to use already developed, efficient solutions
to be evaluated in accordance with the best practice principle for local challenges. Southern European countries have been able to accumulate a wealth of experience in combating vegetation fires with extreme fire behavior for decades. These conclusions must be transferred proportionately to Germany, and especially to the state of Brandenburg, in order to take precautions for the expected next generation of vegetation fires (fires with ever-increasing intensity and increasing risk potential for the population, settlements, and infrastructure) in the North German Plain.

**Proposed solution:** For the orderly coordination and placement of internships and training positions abroad, a central coordination office should be established at the federal level (e.g., at the BBK or the BABZ), but at least in the state of Brandenburg, possibly at the Forest Fire Competence Center. This office can be a

Establish a network of authorities and institutions with foreign BOS, serve to bundle requests and ensure a synergistic and even distribution of places.
At the same time, this office relieves the relevant foreign authorities of single and multiple inquiries.

### 3.6. Central State Procurement and Technology

The central procurement of fire service and disaster protection equipment by the state of Brandenburg has proven successful over the past decades. This has enabled the vehicle fleets of the fire and disaster protection agencies to be significantly rejuvenated and supplemented with modern equipment that reflects the state of the art and meets the required capabilities.
Through centralized tenders combined with the extensive allocation of funding, the special risks of vegetation fires in the state of Brandenburg were taken into account. 56 tank fire engines (TLF 20/50, TLF 5000, and TLF-W Type Brandenburg (TLF-W BB), specially equipped for forest fire operations, have been centrally procured since 2008.
Firefighting vehicle types for municipal fire protection also come from procurement measures of the state of Brandenburg, which since 2020 have also been equipped with special loading kits for vegetation fire fighting to further increase their capabilities.

At the initiative of the LSTE, the DIN Firefighting Vehicle Standards Committee developed a uniform Specification of minimum standards for tank firefighting vehicles for vegetation fire fighting which, since October 2022, as DIN/TS 14530-29, has been demonstrating the technical-tactical operational value of such vehicles in the Federal Republic of Germany and is intended to serve as a starting point for further technical developments.

The forest fire events since 2018 provided an opportunity to further develop the design and equipment of the special tank fire engines for vegetation fire fighting and to initiate a centrally funded large-scale procurement of 35 tank fire engines TLF-W BB, in which
The state of Mecklenburg-Vorpommern also participated with a further 11 vehicles. Some of the vehicles delivered in 2022 have already achieved their high
A further significant capability enhancement was achieved through the procurement of five high-performance conveyor systems (HFS),
consisting of a roll-off container on an all-terrain swap body and a logistics vehicle. These systems enable large-volume water pumping for both fire and flood operations. The necessity of maintaining and the performance of these systems was demonstrated in 2022, among other things, during the major forest fire.
in the Lieberoser Heide. They were able to pump up to 3,000 l/min of fire-fighting water from a lake, over several kilometers are made available for firefighting.

#### 3.6.1 Personal protective equipment

The personal protective equipment (PPE) provided by the municipal authorities is often considered unsuitable by the emergency services for fighting vegetation fires.
described. Often, emergency personnel are only equipped with clothing for indoor firefighting. The outfit for outdoor firefighting is not available to many emergency personnel. As a result, emergency personnel do not wear full or thermally insulating clothing during vegetation firefighting due to the physical strain.

PPE, so that the probability of injuries and physical overload is increasing. The Brandenburg Fire Service Accident Insurance Fund also warns against this fact.

**Proposed solution:** The local authorities should check whether the emergency services

*33for thermal body protection during indoor firefighting (e.g. in apartment fires)* are to be equipped with operational clothing for outdoor firefighting.[34] In this respect, it should be examined whether, in the future, appropriate clothing can be offered at economical prices through framework contracts with the department store of the Central Service of the Brandenburg Police. The addition of a possible additional set of PPE also contributes to prevent the spread of contamination and to improve site hygiene[35] .

**3.6.2. Central State Procurement**

The perceived extreme fire behavior of the forest fires should be understood as a reason to evaluate the technical-tactical approach. Dangerous situations (e.g.
Even with careful tactical selection, sudden, local changes in wind direction cannot be completely avoided. The self-protection capabilities of units must be tested through appropriate technical vehicle and superstructure configuration upgrades, and, if necessary, to evaluate further vehicle types for inclusion in the procurement portfolio.
Selected equipment additions for existing vehicle and logistics components can adequately expand both the special hazard prevention measures for vegetation fires and the supply concepts.

**Solution 1:** The special TLF-W BB are to be evaluate and, if necessary, further develop. Other special vehicle types with a special tactical operational value for major incidents and disasters (e.g. LF-KatS) should be included in the central state funding.

**Proposed Solution 2:** Local fire protection agencies should evaluate their equipment and resources for effective vegetation firefighting and, if necessary, supplement them with the following recommendations. The equipment sets can also be used for the district vegetation fire training program that is currently being developed (see 3.5 Training/Tactics).

- Equipment set D-Storz fittings: The application rates that can be displayed are sufficient for control thresholds of up to 2 m diagonal flame length for direct fire fighting. The fittings (pressure hoses, distributors, jet pipes) should be configured for two groups For transport over rough terrain, transport backpacks/cargo carriers should be used.
- Equipment set hand tools/fire backpacks: attack tools (fire beater, Firefighting tools (e.g., fire backpacks, etc.) are only suitable for direct firefighting in moderate fire intensities (waist-high flames). Hook and digging tools, on the other hand, are suitable for creating control lines/wound strips to defend unaffected areas.
- Circular sprinkler equipment set: Circular sprinklers are suitable for unmanned barrier positions along paths and skid trails, especially in areas contaminated with explosive ordnance. In addition, circular sprinklers can be used to achieve long-term moistening of the forest floor and a beneficial increase in the humidity level under the canopy. The equipment sets should have a coverage range of 500 m and be supplemented by additional fittings (distributors, shut-off valves). Unmanned operation ensures gentle and safe personnel deployment.

*34Clothing should be suitable for vegetation fires, but also for technical assistance*
*35Changing a contaminated PPE set while using the second set*

- Helicopter security equipment kit: Special equipment and PPE are required to secure and operate temporary outstations during airborne support operations. Security equipment includes, among other things, barrier/
Marking material and windsocks. The materials should be sufficient for an area of at least 200 m x 100 m (equivalent to two landing areas for a CH-53, if necessary landing areas for a CH-47F). In addition, special PPE (special
Helmets with communication ports, eye protection, etc.) are required. When assembling this equipment, technical coordination with the FUK BB and the BG Verkehr (Traffic Safety and Health Association) must be ensured.

- Accommodation equipment kit: For self-sufficient ad-hoc accommodation of personnel of the disaster control units (here: BSE, leadership, ICT), accommodation materials should be (sleeping bags, folding beds, hygiene products, towels, ...) for at least 100 people. The set should at least be sufficient for accommodation in structural
Facilities (gymnasiums, multi-purpose halls, etc.) can be configured. If necessary, a
The addition of tents, heating, power supply and lighting should be evaluated.
In addition to a multi-day (at least 48-hour) deployment of disaster control units in the event of vegetation fires, these equipment sets could also be used for Brandenburg relief contingents in more remote operational areas.

- Hygiene equipment kit: Vegetation fires often occur far from settlements without adequate infrastructure. Roll containers can be set up at catering and logistics points for rough cleaning of contaminated skin.
Mobile toilets, which are also procured to meet human needs, are delivered promptly to these points. Depending on the staff size of a BSE, six toilets, two urinals, and two washbasins should be installed (see user key according to ASR 4.1).

### 3.6.3 Involvement of agricultural enterprises

Agricultural businesses and contractors provide valuable and appreciated support in containing vegetation fires. This is due not least to the agricultural and forestry value creation threatened by fires. Water transport and soil cultivation technology can initiate rapid and comprehensive measures. In the event of large forest fires, numerous offers of assistance from agriculture reach the responsible emergency response agencies. Due to priority decision-making and organizational needs, as well as the nature of the individual requests, a coordinated response to the offers could not be provided in every case. In addition, there were isolated cases of uncertainty regarding possible financial demands that were not communicated during the requests, which remained after the end of the operation.

At the same time, harvesting operations on agricultural land are a potential cause of fire. Rotating machine parts and foreign objects can represent ignition sources in dry grain fields. Incipient fires can spread if safety measures are inadequate.
and lead to the endangerment of agricultural machinery and the loss of crops. Rapid fire escalation to adjacent forest and settlement areas must be
should be considered in the risk analysis. Due to the wind-driven nature of the open terrain, the fire front's travel speeds are generally assumed to be very high.

**Solution 1:** For targeted planning availability and better overview-

The agricultural technology potential, such as water barrels, tractors
Using specialized technology, local emergency response authorities can compile the information in an online database prior to an incident. Voluntary registration allows for transparent selection and orderly integration by decision-making bodies, regionally and based on capabilities. Agricultural interest groups (Brandenburg State Farmers' Association) and fire service organizations (Brandenburg State Fire Service Association) could be involved in disseminating the concept or inspiring its adoption. One example is the "Red Farmer" concept of the KFV Main-Spessart eV36 .

**Proposed solution 2:** Farmers must be further sensitized to preventive measures during harvesting on dry grain fields. The "Regulatory Ordinance on the Installation of Weed Strips on Agricultural Land at the Edge of Forests" of February 23, 2021, provides a good basis for campaigns that can be designed jointly with the interest groups of all involved stakeholders (the Baden-Württemberg State Fire Brigade Association, the Baden-Württemberg State Farmers' Association, and the Baden-Württemberg Forest Owners' Association).

### 3.7. Preparation for deployment

The basic risk of forest fires in the state of Brandenburg is regularly assessed as high to very high during the forest fire season. This basic assumption will change in the coming years.
and decades, but will probably become even more pronounced. In this context, it is necessary for high-risk forest fire hotspots (e.g. poor or deteriorating firefighting water supply, settlement areas adjacent to forests, poorly

developed road system, suspicion of explosive ordnance) at local and supra-local level. Legally required local operational plans must contain basic procedures for the deployment of forces and resources, assembly points

and adaptable leadership and communication structures are to be defined together with the responsible forestry authority and are reflected in the special deployment plans of the districts.
If forest fire hotspots extend over the territory of several local authorities, homogeneous coordination must be achieved through inter-municipal cooperation.
Disaster management authorities support this process through uniform specifications (e.g., keyword-based force/resource approaches) as well as through the creation and updating of event-specific special plans, which should be synchronized with other specialist authorities (e.g., forestry authorities, mining authorities, explosive ordnance disposal services). The special plans represent
represent an important organisational support documentation of measures and available special funds and thus complement a preparation level that goes beyond the local level
Planning of operational resources within the framework of supra-regional fire protection. The quality level of the preparatory actions of the aforementioned responsible bodies varies within the state of Brandenburg.

Furthermore, retrospective conclusions beyond the collection of forest fire data are important for deriving appropriate and well-founded adaptation strategies for the prevention and response to vegetation fires in the future. This must also include fires on agricultural and other open land areas.36

*36 https:// kfv-msp.de/ redfarmer/*

### 3.7.1. Alarm and deployment order/force-resource approach

Intermunicipal cooperation has been strengthened through central state procurement since 2007 provided a necessary impetus. Many local authorities have since harmonized their alarm and dispatch procedures to ensure appropriate emergency response. For vegetation fires, alarm and dispatch procedures are also coordinated between the relevant authorities, differentiated regionally. To ensure a future opportunity to limit the spread of a vegetation fire with extreme fire behavior, even in the initial phase, further developed minimum requirements for an adequate response should be recommended or defined. A discussion paper on critical vegetation fires

was published by the LSTE in the trade journal BRANDSchutz/Deutsche Feuerwehrzeitung in the Issue 09/2022 was published to create a standardization impulse for dynamic spreading situations, such as those to be expected in vegetation fires under certain conditions. Even if the local alarm and dispatch regulations are individually implemented within the framework of municipal self-government, uniform and effective
Recommendations for appropriate initial alerting by the responsible authorities for supra-local fire protection and supra-local assistance in relation to the respective local risk potentials are considered effective. Some local alarm and dispatch regulations for vegetation fire-specific operational keywords are considered a minimal approach, although the level of availability of suitable, alertable emergency resources in the state of Brandenburg has increased significantly in recent decades.

**Proposed solution:** To contain a dynamic vegetation fire, the following preliminary planning is recommended for each district. The required extinguishing water volumes via tank fire engines (TLF) should be planned in advance, depending on keywords and local needs. As a first indication, at least 12,000 liters of water may be appropriate for the initial alarm in peripheral areas. In addition, additional fire engines in squadron or group strength, as well as a command unit to establish a growing command organization, are necessary. 16 emergency personnel with 6,000 liters of extinguishing water (via TLF) for early intervention (usually 25 minutes after the fire is reported) at a priority fire flank appear appropriate. Additional personnel and resources

must be considered for the second flank or for the protection of settlement areas.
If the local fire protection authority cannot manage the described functions and resources with its own resources, intermunicipal cooperation must be initiated to coordinate the alarm and dispatch regulations. The supra-local fire protection authority must provide guidance and assume a coordinating role while maintaining basic local protection. For highly urban landscapes without large forest areas, reduced approaches may be sufficient.

The Working Group on Operational Keywords is currently revising the operational keyword catalogue of the Integrated Regional Control Centres.

### 3.7.2 Analysis of vegetation fires and weather data

Qualified data sets on vegetation fires in the state of Brandenburg are currently limited exclusively to fires that caused damage to forest areas. Fires in grain fields and other unpopulated open areas, however, are not included.
centrally recorded. This reduces the retrospective, central significance of actual ecological

economic and ecological damages and risks as well as the strain on forces and resources. Vegetation fires place far greater demands on emergency services than the annual forest fire statistics of the Brandenburg State Forestry Authority reflect. Grain field and other open land fires pose an equal risk, at least in the summer months, since Preventive protective measures have so far appeared to be of little need. However, grain fields often border directly on residential properties with flammable building materials and can endanger entire residential areas in the event of a fire. Several near misses demonstrate also in the state of Brandenburg (e.g. Wildgrube 2019 in EE, Prösen 2020 in EE, Lindenberg 2021 in LOS, Lossow 2021 + 2022 in FF, Zossen 2022 in TF) a perceptible, but currently Unfounded risk for settlement areas in the case of dynamic and fast-moving field fires. The speed of movement and the fire load-controlled fire intensity, including the prevailing weather parameters (humidity, wind speed, and temperature), of field fires also result in special safety regulations for emergency responders. However, general preventive measures can only be drawn in a balanced and transparent manner in the interest of all affected parties (emergency responders, farmers, landowners) if objective data and fire progression analyses are available for evaluation.

Furthermore, fire progression analyses, especially in the case of forest fires, can be used to derive general patterns of spread and behavior for similar future situations. An investigative, qualified analysis of forest fire progressions is currently underway. (At least) not in an interdisciplinary manner. From these scientifically oriented follow-up analyses, important conclusions can be drawn for preventive, as well as defensive, tactical measures. In particular, the situational conditions on site, such as the vegetation condition, profile incisions, and local weather and wind data, are important data sources for constructive analysis.

Location-specific and precise weather data will play an even more crucial role than before in the future assessment and analysis of fire behavior and in the selection of tactics. In particular, local winds, solar intensities and topography are taken into account in addition to the relative Humidity, outside temperature and residual moisture in dead wood, litter and The raw humus layer on site represents a significant assessment factor for determining operational priorities. In southern Europe, special units with expertise in forestry, meteorology, and hazard prevention have been established to exclusively produce spread forecasts for extreme vegetation fires in order to enable the precise deployment of forces and resources.

**Proposed solution:** The statistical survey at a central location must cover the entire spectrum of vegetation fires and must not be limited to forest fires. It is expected that the field of fire analysis will not only become increasingly relevant for ad hoc evaluation during an operation, but will also play a key role in the collection of data sets for strategic adaptation (preventive and defensive vegetation fire measures) in an interdisciplinary context.

### 3.7.3. Specialized vegetation firefighting units

The extreme vegetation fires of the 2022 season have demonstrated that firefighting requires specialized expertise in situation assessment and technical-tactical measures. This expertise, which needs to be further developed, leads to special skills which cannot be covered by every local fire department in the state of Brandenburg. Overburdening these local units must be avoided given the diverse range of operational scenarios. Local fire departments are to be viewed as general intervention units within the framework of general basic protection. The specific capability development for fighting and containing large forest fires that are difficult to control requires a pooling of personnel and material resources to complement local fire department structures. Given the lengthy process of acquiring skills and experience, it seems sensible to establish units analogous to non-profit organizations such as

@fire International Disaster Relief Germany eV or Forest Fire Team eV in the
To specialize and consolidate fire and disaster protection structures. As a
A positive model example from the Oberhavel district shows that inter-municipal cooperation can efficiently facilitate the transfer of international knowledge and practical skills.
The fire departments of the towns of Liebenwalde and Zehdenick established an intermunicipal vegetation firefighting unit (SKV - Specialized Vegetation Fire Forces).
and conduct harmonized training events and procurement projects to supplement the unit's technical capabilities. Meanwhile, the city of Templin (UM) has also joined this
connected to a specialized unit.

The aforementioned organizations and units were represented in some of the
The company has been successfully deployed in complex forest fire incidents (Beelitz, Bad Liebenwerda/Mühlberg, Falkenberg). This was due not least to its special expertise in tactical firefighting operations as well as in structured and qualified follow-up firefighting operations.

**Proposed solution:** With the expected further increase in extreme vegetation fires, it can be assumed in the medium term that this non-governmental organization (NGO) expertise will not always be available on a regular basis. Equivalent alternatives must be developed for these situations. These include the intermunicipal, contractual merger of interested emergency services into specialized units or the establishment of special units at the level of the endangered districts.

### 3.8. Preventive forest fire protection

The defensive measures for forest fires are essentially a short-term treatment of symptoms. The causes of fire initiation and progression, however, are very complex and multiple. Although the progression of a vegetation fire can be influenced by silvicultural and other preventive measures, the
The causes of fire outbreaks are difficult to model. The majority of forest fires are caused by negligent or intentional anthropological triggers. While vegetation fires far from settlement structures primarily affect agricultural and forestry areas,
and cause ecological damage, the risk situation of fire
vegetation area near settlement areas is significantly larger. This is
particularly due to site-specific development factors such as highly flammable
Vegetation, its vertical and horizontal density, deadwood components, and the actual distance from settlements to forest areas. In the presence of unfavorable weather and drought factors that cannot be influenced at the same time, fire events can no longer be controlled with conventional means of hazard prevention, even if the fire load is sufficient.

In such situations, tactical defensive measures will in future be less focused on the

the actual liquidation of the fire, but rather on the defense of the settlement areas
According to long-term forecast trends, an intensification of extreme fire behavior must be assumed, making preventive measures around settlements and infrastructure even more important than they already are.

### 3.8.1. Fire load management

The discussion about deadwood as a fire load in forests is currently controversial. Calamity wood37 can indeed pose a major challenge for emergency responders during an extensive vegetation fire. Even deadwood in contact with the ground, which dries out during a long drought period, can lead to fire intensities with above-average heat release rates. Firefighters perceived the spread and propagation speeds in the 2022 forest fire season as very high due to the dry deadwood.
If forest damage and tree mortality due to multiple factors continue, the fire load from dry deadwood must be expected to continue to increase. Deadwood plays an important ecological role in the forest ecosystem, particularly in the silvicultural conversion process from fire-prone coniferous forests to more climate- and fire-resistant mixed forests. Preserving deadwood on burned areas serves to protect
and promoting forest regeneration, e.g. through shading, buffering of temperature extremes, wind protection and water retention, maintaining the highest possible nutrient stocks and promoting biodiversity. Compromises must be found here. Decisions regarding the handling of deadwood, particularly those caused by fire, must therefore be made on a small scale and take into account the proximity to paths, settlements and other
Infrastructures in the sense of preventive fire protection. On forest paths that are
Clearing deadwood is a top priority for forest fire and disaster protection. The permanent accessibility of these roads must be ensured. The brochure "Recommendations for Managing Forest Fire Areas" provides further clear recommendations for dealing with deadwood. These recommendations are aimed at all forest owners and serve as a basis for forestry experts to advise, monitor, and implement measures.

Furthermore, the system of forest fire protection strips and fire lanes is an important preventive measure for suppressing fire propagation along road and rail infrastructure. Some fire lane systems have not been maintained in recent years.
and lost their effectiveness. The establishment of forest fire protection strips and fire protection belts should be encouraged again, particularly along federal and state roads and some railway lines. Grass growth in forest areas below a light canopy closure level can promote the spread of fire and should therefore be avoided.

**Proposed solution:** To protect forest edges along settlement areas and forest paths as
To be able to use these defensive lines, a compromise solution could be the removal (chop) of deadwood and low branches to the left and right of forest paths (50 m each). A parceling of forest areas to minimize fire spread is proposed.
In the long term, the most sustainable goal is to convert forests to mixed forests along settlement areas adjacent to forests. Furthermore, the inner and outer edges of forests are to be
To establish a protective function and promote its development.

*37Wood resulting from storm damage, drought and/ or pest infestation*
*38MLUK: Recommendations for managing wildfire areas, December 2022*

Furthermore, in suitable forest and open land areas, the forestry authorities should also examine whether fuel and its supporting conditions can be reduced through targeted maintenance fire measures (burning out weedy layers) (preventive

fire operation).

#### 3.8.2. Spatial planning

The transition zones between potentially fire-prone vegetation areas (even seasonally) and settlement areas must be subject to special consideration. This so-called wildland-urban interface management will particularly affect settlements (forest settlements, weekend cottage settlements, campsites) that are directly adjacent to or surrounded by forests. This management may also include

Agricultural areas should be included, as a fire outbreak (harvest-ready grain fields with low residual moisture) on these open areas is expected to spread rapidly. Grain field fires with dangerous spread potential in the 2022 season support this assumption (e.g., the city of Zossen, the Verbandsgemeinde Liebenwerda/

Fichtenberg municipality/border to Saxony). The urban development framework conditions are adapted to the risks arising from vegetation fires near settlement areas
Due to the expected increase in extreme fires resulting from multiple, location-dependent influencing factors, sensitive considerations must be prioritized in the future designation of building sites and the approval of construction projects in these areas, which excludes any danger to buildings and users or at least reduces it to an acceptable level.

For existing structures that are surrounded by or border on forest areas, authorities must be able to implement subsequent risk-minimizing measures. It is important to increase the protection of citizens in these settlement areas and to implement special,
to allow preparatory defensive measures that go beyond the usual scope.

**Proposed solution:** Planning authorities should already consider the risks of vegetation fires in the future during the development planning phase. Wildland-Urban Interface Management not only focuses on reducing fire loads, but also aims to plant alternative, more fire-resistant tree species (creating forest fire barriers). Especially when designating new building areas within the framework of development planning, distances and protective measures must already be taken into account.

The design specifications for construction works to be erected in the immediate vicinity of vegetation areas must take into account the latent risk of spreading
of a vegetation fire on these very structures. Furthermore, the storage of combustible materials and the planting of highly flammable trees and shrubs in the immediate vicinity of buildings should be prohibited in individual cases. Fire bridges and fuel clusters (e.g., gutters filled with needle litter, roofing felt) on properties provide favorable conditions for fire spread to building structures.

### 3.9 Cross-border cooperation

In the initial phases of two large forest fires involving vegetation fires in the area of the Saxon-Brandenburg state border (Verbandsgemeinde Liebenwerda), the direct and indirect communication coordination processes were assessed as being in need of further optimization.

In particular, operational measures (technical-tactical possibilities) and operational objectives were not permanently defined in the same way on the Brandenburg and Saxon sides and were
Not congruent. Only by installing liaison officers and relocating mobile command posts could an improvement be achieved. Furthermore, hurdles in technical communication (use and operation of call groups in the BOS digital radio system) were identified.

On the one hand, this is due to the differentiated design of the respective fire and disaster protection laws. On the other hand, pre-coordinated operational procedures along state borders are not the rule, but rather the exception (for example, in the Spree-Neisse district). The occurrence of a vegetation fire along the state border

with the risk of spreading to the other federal state appears possible at any time and is not limited to the border area between the Free State of Saxony and the state
Brandenburg. The latent risk encompasses all federal states bordering Brandenburg with correspondingly high forest fire risk classes.

#### 3.9.1 Deployment of advance teams

Even in the early stages of deployment, when responsibility still lies with the local authorities, coordination between neighboring authorities along territorial boundaries appears to be regularly necessary. While this has been coordinated organizationally in advance and is now a common practice within the state of Brandenburg, it has not yet been established to a comparable extent across the state's borders.

The local exchange of information between two responsible bodies is the simplest
and unbureaucratic solution without calling into question the respective local jurisdiction.
When situations escalate, the mutual deployment of liaison officers can ensure the mutual exchange of situation information and the coordination of operational measures.

When deploying contingents to another federal state, it has become established practice to send an advance command to assess the situation and prepare for deployment.

**Proposed solution:** Local and supra-local authorities should jointly develop mission-specific and, if possible, uniform procedures depending on different operational scenarios and threat levels. Within the established command organization, suitable contact points must be established, from situational awareness transfer to harmonized procedures. Both technical and organizational synchronization processes are

to describe this. The personnel composition should include liaison officers and command units as a type of advance and follow-up command. Examples of concepts proposed by the municipality of Zeithain (Meissen district, SN) and the Verbandsgemeinde Liebenwerda (EE) are worth mentioning here.

A regular, joint exchange of experiences between the federal states bordering Brandenburg (focus: BB-SN, BB-ST) to present the legal
This approach can complement the framework conditions in order to not only create understanding but also to identify and resolve technical and organizational challenges before the next cross-border deployment.

# Conclusion 4.

Since the 2018 season at the latest, major forest fire events have led to a sensitization
While it was initially assumed that this was a one-off anomaly in the fire service, the conclusion of the 2022 forest fire season is clear: the hazard potential posed by vegetation bands has not diminished.
Rather, an increasing trend can be observed. In no other forest fire season in the history of the state of Brandenburg were more settlement areas and their residents endangered than in
The country's fire and disaster protection system, which relies primarily on volunteers, briefly reached a capacity limit in June and July 2022 that was difficult to compensate for.

This trend, which has been subjectively observed in the state of Brandenburg since 2018, gives an idea of the challenges that emergency response authorities may face in the coming years and decades. The 2022 report of the United Nations Environment Programme39 also represents a further opportunity to raise awareness for the state of Brandenburg.
The short-term risk of extreme vegetation fires will increase worldwide by a factor of 1.08 to 1.14 within the current decade. By the turn of the century, this risk factor is forecast to be between 1.36 and 1.57. However, a global forecast can only be differentiated based on local climate development and landscape and natural area design.
can be transferred, the risky complexities of the state of Brandenburg will tend not to be reduced by the previously dominant pine stands and large, contiguous agricultural areas.

Every preventive measure to minimize hazards is more sustainable than further optimization of defensive processes and capabilities. However, preventive measures, ranging from the creation of protective zones around settlements and the clearance of explosive ordnance to the conversion of climate- and fire-resistant forests, are to be viewed as a lengthy and, in some cases, a generational task. The perceptions of an increasingly extreme
Fire behavior is site-specific due to the interrelationships between forest, agricultural, and residential areas and is a defining feature of the landscape design in the state of Brandenburg.
Disentanglement and compensation measures will also be implemented over decades.
Claim.

At least during the long-term implementation phases of preventive measures, additional efforts in the continued development of defensive forest fire protection are desirable, taking into account a persistent or further increasing risk.
Not only investments in the provision of special technology appear necessary to minimize the direct impacts of vegetation fires. Improvements in setup and operational processes during acute deployment phases can contribute to minimizing devastating personal injury and property damage, as can special capability adaptation.
Specialized skills acquisition within the framework of training and further education is of particular importance. Only when specialized technology can be integrated into operational tactics and capability development in a situation-appropriate manner can significant increases in effectiveness be achieved. This also requires interagency synchronization processes to ensure that forest fire protection, in the interests of both population and nature conservation, is a common

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*39United Nations Environment Programme (2022). Spreading like Wildfire – The Rising Threat of Extraordina-ry Landscape Fires. A UNEP Rapid Response Assessment. Nairobi.: https:// wedocs.unep.org/ bitstream/ hand-le/ 20.500.11822/38372/ wildfire_RRA.pdf (Zugriff: 15.12.2022)*

The task must be viewed without rigid jurisdictional boundaries. Only through appropriate compromises can risks be reduced, but not completely eliminated. Intensive awareness-raising in the wider societal context and an interdisciplinary understanding must become the pillars of sustainable vegetation fire prevention and control.

The multitude of actors involved from the various BOS and specialist authorities in
The past forest fire season has shown how important it will be to concentrate specialist personnel in prevention and operational preparation.

In addition, the fields of action described in the previous chapter must be continuously addressed and evaluated.

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